EE361 Electromechanical Energy Conversion I — METU EEE — full-semester study notes

Chapter IV — Electromechanical Energy Converters Basics and DC Machines

Elektromekanik Enerji Dönüştürücülerin Temelleri; DA Makineleri · Weeks 10–12 of the 2026 syllabus

- Lesson 22 — Principles of electromechanical energy conversion: Lorentz force, energy balance, energy and coenergy
- Lesson 23 — Force calculation in linear electromagnetic systems; magnetic stress
- Lesson 24 — Torque in rotating systems, reluctance motors, multiply excited systems and dynamic equations
- Lesson 25 — DC machines: construction, commutation, generated EMF and electromagnetic torque
- Lesson 26 — DC machine equivalent circuits, power flow, and separately excited generators and motors
- Chapter IV — DC-machine archive problems with full solutions

> Assistant-written teaching notes (English with Turkish terminology). Teacher sources LN12–LN15; textbook Sen Ch. 3–4. Appendix: DC-machine archive problems with full solutions. Every numerical result is checked by `assistant/calculations/full-semester/run_all.py`; source pages are cited in each lesson. Printed 2026-11-02 from the Markdown sources in `assistant/lessons/`.

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1. Final Fall 2017 Q2 Part II — the simplest DC machine
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5. Final Fall 2018 Q3 — eDumper electric mining truck
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8. Jan 2023 Final Q2 — moving core: force and induced voltage along a motion profile
9. Jan 2023 Final Q3 — inductor and brushed DC machine structure
10. Jan 2023 Final Q4(a)–(c) — crane motor with armature and field control
11. Jan 2022 Final Q3 Part II — PM DC wind generator charging a battery

Lesson 22 — Principles of electromechanical energy conversion: Lorentz force, energy balance, energy and coenergy

Chapter IV: Electromechanical Energy Converters Basics (Elektromekanik Enerji Dönüştürücülerin Temelleri) · Week 10 · Lesson 22 of 31

Assistant-written teaching note. Teacher source: LN12 slides 2–16 (Lorentz force, losses, lossless system, energy balance, energy and coenergy). Book: Sen §3.1–3.2.1 (printed pp. 93–99, viewer 117–123), Examples 3.1–3.2; §4.1 (pp. 119–121) motional voltage and force on a conductor. Numbers: `lesson22_energy_coenergy.py`.

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1. Place and goals

Chapters II and III dealt with magnetic systems in which nothing moves; energy entered and left only through electrical terminals. Motors, generators, relays and actuators add a **mechanical port**: a part that moves under magnetic force. Chapter IV asks two questions: *how large is the force or torque?* and *how does electrical energy become mechanical energy (or the reverse)?* This lesson builds the two tools: the **Lorentz force** on currents, and the **energy method** based on the field energy and coenergy of Lesson 14. Lesson 23 uses them for forces in linear systems, Lesson 24 for torque in rotating systems, and Lessons 25–26 for DC machines.

You should be able to:

1. Compute the force on a current-carrying conductor ($\mathbf{F} = I\mathbf{l} \times \mathbf{B}$) and the torque on a simple rotor, with directions from the right-hand rule.
2. Write the energy balance of an electromechanical converter, identify its losses, and reduce it to a lossless coupling field with one electrical and one mechanical port.
3. Define field energy and coenergy on the λ–i plane, and explain why their sum is λi.
4. Derive the force from energy at constant flux linkage and from coenergy at constant current, and show the two agree.
5. Explain where the mechanical work comes from in slow (constant-current) and fast (constant-flux) motion.

Prerequisites: Lesson 14 (stored energy, coenergy), Lesson 15 ($e = d\lambda/dt$).

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2. The Lorentz force

2.1 From a charge to a conductor

A charge q moving with velocity $\mathbf{v}$ in fields $\mathbf{E}$, $\mathbf{B}$ experiences $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$; in a purely magnetic system $\mathbf{F} = q\,\mathbf{v} \times \mathbf{B}$ (LN12 slide 2). Summing over the moving charges in a volume gives the force density $\mathbf{f} = \rho\mathbf{v} \times \mathbf{B} = \mathbf{J} \times \mathbf{B}$ (N/m³). Integrating over a straight conductor of length l carrying current I in a uniform field (LN15 slide 10):

$$\mathbf{F} = I\,\mathbf{l} \times \mathbf{B}, \qquad |F| = BIl \ \text{ when } \mathbf{l} \perp \mathbf{B} \qquad \text{(Sen Eq. 4.2)}. \tag{22.1}$$

The direction follows the right-hand rule of LN12 slide 3 (current along the thumb, field along the fingers, force out of the palm), or equivalently the cross product. LN12 slide 4 lists applications: homopolar motors, electromagnetic aircraft launchers, railguns.

2.2 The companion: motional voltage

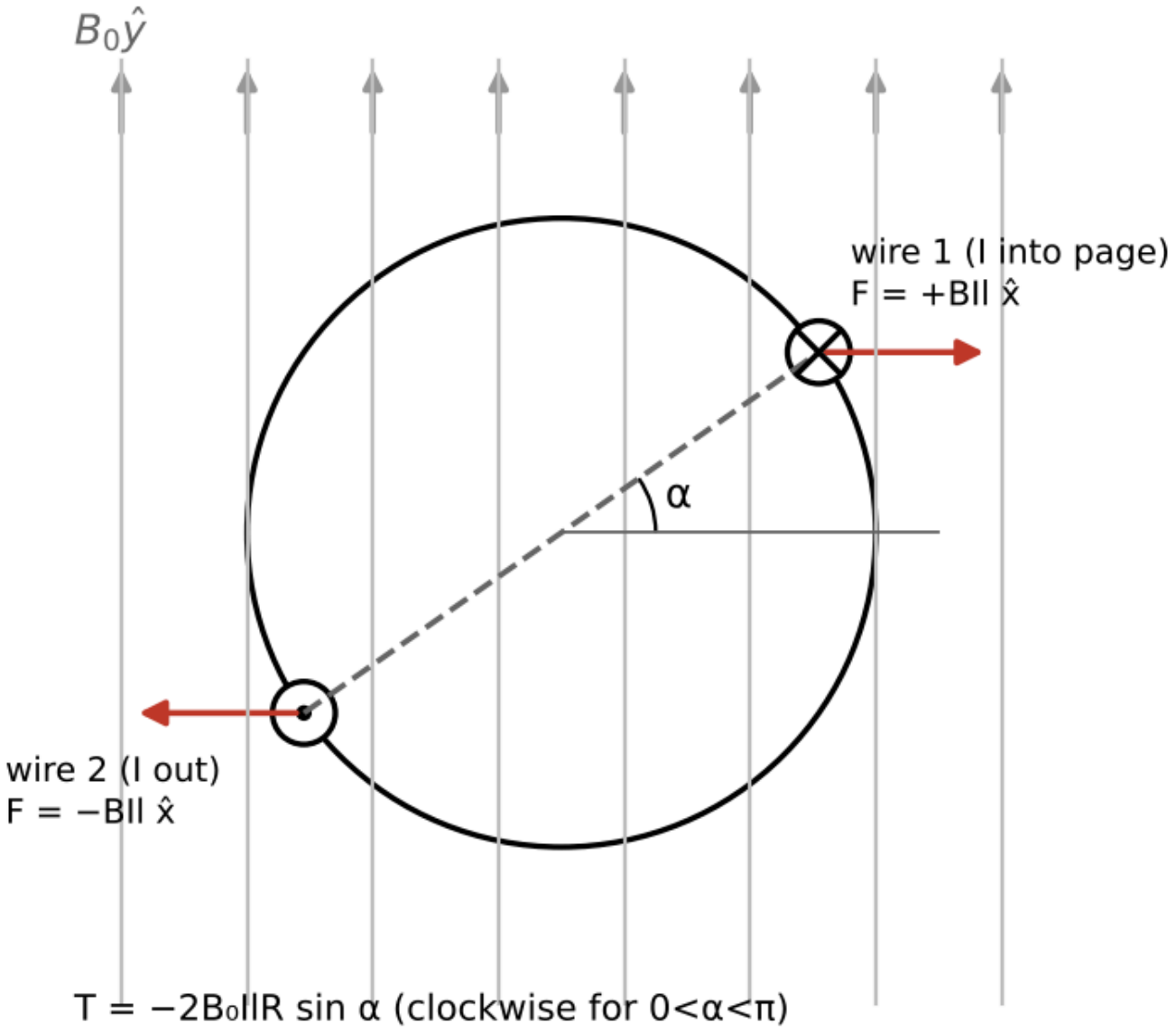
A conductor moving with velocity $\mathbf{v}$ through $\mathbf{B}$ has an induced EMF $e = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l}$, i.e. $e = Blv$ for mutually perpendicular quantities (Sen Eq. 4.1). Force and motional EMF always appear together: if a conductor carrying I moves at speed v, the mechanical power $Fv = BIl\,v$ equals the electrical power $eI = Blv\,I$. This one identity is the heart of every DC machine (Lesson 25).

2.3 Torque on a rotor in a uniform field (LN12 slide 5; LN15 slide 8)

Two conductors of length l sit diametrically opposite on a rotor of radius R in a uniform field $B_0\hat{y}$; wire 1 at angle α carries I into the page, wire 2 carries I out (Figure 22.1). Wire 1: $\mathbf{F}_1 = I(-l\hat{z}) \times B_0\hat{y} = +B_0 Il\,\hat{x}$. Wire 2: $\mathbf{F}_2 = -B_0 Il\,\hat{x}$. Torques about the axis, $\mathbf{r} \times \mathbf{F}$, add:

$$T_z = -2B_0 IlR \sin\alpha. \tag{22.2}$$

For $0 < \alpha < \pi$ the torque is clockwise and drives the rotor toward α = 0, where it stops (stable equilibrium). That answers LN12's "determine the direction of rotation" and shows why a simple coil cannot rotate continuously: the current must be reversed every half turn — the job of the **commutator** (Lesson 25).



2.4 When the Lorentz force is not enough

$\mathbf{J} \times \mathbf{B}$ gives the force on *currents*. It does not directly give the force on *iron*, which carries no current — yet an electromagnet lifts scrap, and LN12 slide 6 asks what happens to a separate iron bar placed across the gap of an excited C-core: it is pulled in to close the gap. This **reluctance force** (relüktans kuvveti) arises from magnetization currents inside the iron and is most easily computed with the energy method below. The energy method works for both kinds of force.

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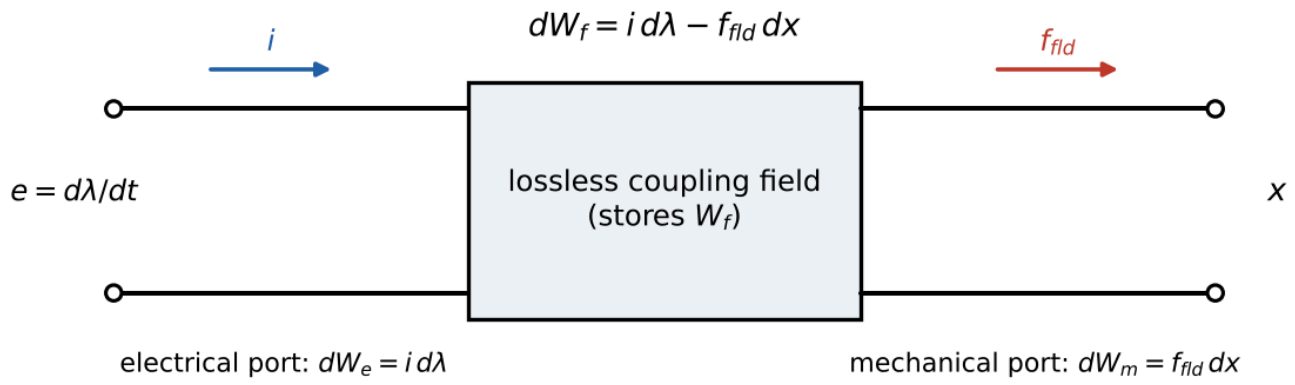
3. Energy balance of an electromechanical converter

3.1 Losses and the lossless core of the problem

Every converter has three systems — electrical, magnetic (coupling field) and mechanical — and three kinds of loss (LN12 slide 8): ohmic (i^2R), magnetic (hysteresis and eddy) and mechanical (friction, windage). Sen Eq. 3.2:

$$\begin{pmatrix}\text{electrical input}\\ -\,i^2R\ \text{loss}\end{pmatrix} = \begin{pmatrix}\text{mechanical output}\\ +\ \text{friction, windage}\end{pmatrix} + \begin{pmatrix}\text{increase in stored}\\ \text{field energy} + \text{core loss}\end{pmatrix}.$$

The losses do not take part in the conversion and can be handled separately (resistors in the electrical circuit, friction torque on the shaft). What remains is a **lossless coupling field** with two ports (LN12 slide 9; Figure 22.2): an electrical port with current i and flux linkage λ, and a mechanical port with force f_{fld} (on the moving part, in the direction of increasing x) and position x.



3.2 The differential energy balance

In time dt the electrical port delivers $dW_e = ei\,dt = i\,d\lambda$ (using $e = d\lambda/dt$), the field does mechanical work $dW_m = f_{fld}\,dx$ on the moving part, and the rest is stored:

$$\boxed{dW_f = i\,d\lambda - f_{fld}\,dx} \qquad \text{(LN12 slide 11; Sen Eqs. 3.3–3.8).} \tag{22.3}$$

In power form (LN12 slide 12): $\dfrac{dW_f}{dt} = i\dfrac{d\lambda}{dt} - f_{fld}\dfrac{dx}{dt}$. The verification script checks this pointwise along a simulated motion.

Key consequence: the stored energy is a state function of (λ, x). Because the field is lossless, W_f depends only on the present flux linkage and position, not on how they were reached. We may therefore compute $W_f(\lambda, x)$ along the easiest path: hold x fixed ($dx = 0$, no mechanical work) and raise λ from zero:

$$W_f(\lambda, x) = \int_0^{\lambda} i(\lambda', x)\,d\lambda'. \tag{22.4}$$

(LN12 slide 14; Sen Eq. 3.9.) This is exactly the stored energy of Lesson 14, now with the position as a parameter.

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4. Energy and coenergy

4.1 Definitions on the λ–i plane

For a fixed position x, the λ–i curve of the coil splits the rectangle $\lambda \times i$ into two areas (LN12 slide 13; Figure 14.4):

- **field energy** $W_f = \int_0^{\lambda} i\,d\lambda$ — the area between the curve and the λ axis;
- **coenergy** $W'_f = \int_0^{i} \lambda\,di$ — the area between the curve and the i axis;

$$W_f + W_f' = \lambda i \qquad \text{(LN12 slide 15; Sen Eq. 3.22).} \tag{22.5}$$

For a linear coil, $\lambda = L(x)i$ and $W_f = W_f' = \tfrac{1}{2}L(x)i^2 = \lambda^2/(2L(x))$. With saturation $W_f' > W_f$. Sen notes that coenergy "has no physical significance" by itself; it is a mathematical tool — but an extremely convenient one.

4.2 Coenergy as a function of (i, x)

Differentiate (22.5): $dW_f' = \lambda\, di + i\, d\lambda - dW_f$. Substituting (22.3):

$$dW_f' = \lambda\, di + f_{fld}\, dx. \tag{22.6}$$

So the coenergy is naturally a function of current and position, just as energy is of flux linkage and position.

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5. Force from energy and from coenergy

5.1 The two formulas

Compare (22.3) with the total differential $dW_f = \dfrac{\partial W_f}{\partial \lambda} d\lambda + \dfrac{\partial W_f}{\partial x} dx$; and (22.6) with $dW_f' = \dfrac{\partial W_f'}{\partial i} di + \dfrac{\partial W_f'}{\partial x} dx$. Matching coefficients:

$$\boxed{f_{fld} = -\left.\frac{\partial W_f(\lambda, x)}{\partial x}\right|_{\lambda\ \text{const}} = +\left.\frac{\partial W_f'(i, x)}{\partial x}\right|_{i\ \text{const}}} \tag{22.7}$$

(LN12 slides 22, 25–26; Sen Eqs. 3.26–3.27; with Φ and $\mathcal{F} = Ni$ in place of λ and i as LN12 writes them.) Also $i = \partial W_f/\partial\lambda$ and $\lambda = \partial W_f'/\partial i$. The two forms give the **same force** at the same operating point; choose whichever matches the given variables (Sen Example 3.2: −124.7 N both ways, Example 22.3).

The phrases "constant λ" and "constant i" describe which variable is held fixed **in the partial derivative**, not the physical conditions of a real motion. The force at an instant depends only on the state (λ, x) or equivalently (i, x).

5.2 Linear systems: one formula to remember

With $\lambda = L(x)i$, $W_f' = \tfrac{1}{2}L(x)i^2$, so

$$f_{fld} = \frac{1}{2} i^2 \frac{dL(x)}{dx} = \frac{\lambda^2}{2L(x)^2}\frac{dL}{dx} \qquad \text{(LN12 slide 22; Sen Eqs. 3.32, 3.34).} \tag{22.8}$$

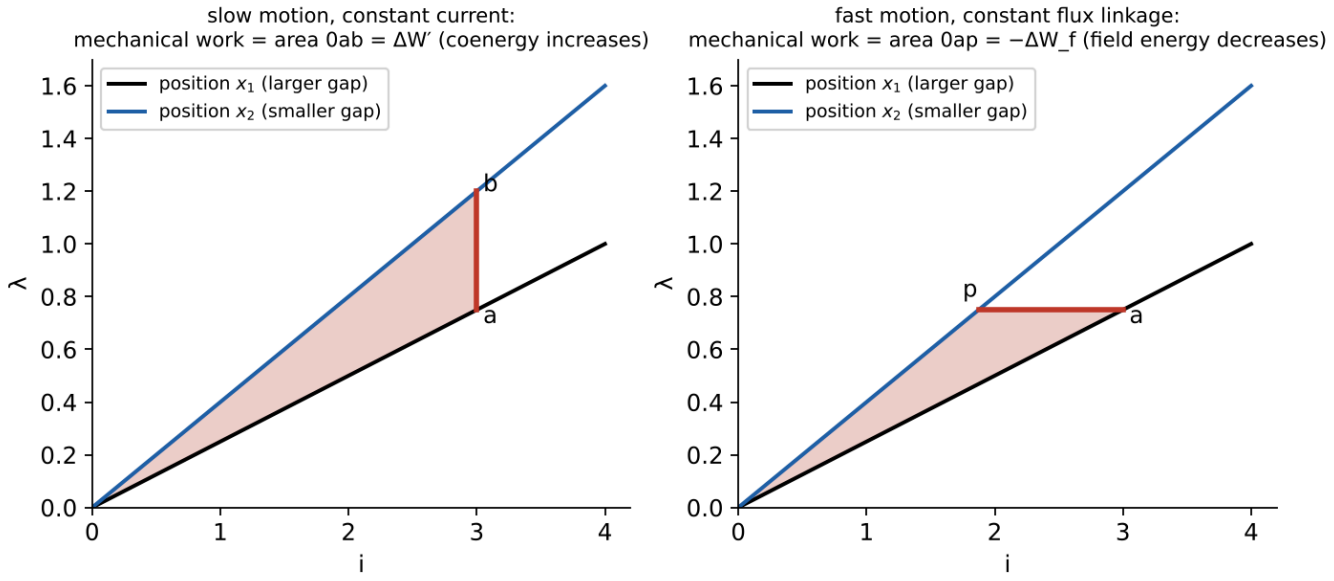
The force acts in the direction that **increases inductance** — equivalently decreases reluctance ($L = N^2/\mathcal{R}$). In LN12's summary (slide 36): "a magnetic circuit tries to reduce W_{mag} at constant flux, i.e. to maximize inductance, to minimize reluctance." Iron is pulled into gaps; gaps try to close; rotors turn toward alignment.

5.3 Where does the mechanical work come from? (Sen §3.3, Figure 22.3)

Let a movable part move from x_1 to a position x_2 with smaller gap (larger L).

- **Slow motion, constant current** (the source holds i): the operating point moves vertically up from a to b on the λ–i plane. Electrical input $= i\,\Delta\lambda$ (rectangle abcd); field energy rises; **mechanical work = area 0ab = increase of coenergy**. For a linear system exactly half of the electrical input becomes mechanical work and half is stored (Example 22.4).
- **Fast motion, constant flux linkage** (the current cannot change fast enough): the point moves horizontally from a to p. No electrical input ($i\,d\lambda = 0$); **mechanical work = area 0ap = decrease of field energy** — the field supplies the work from its store.

For a differential displacement the two areas coincide to first order, confirming that (22.7) gives one force regardless of path.



6. Methods and common mistakes

1. Describe the magnetic circuit as a function of position: $\mathcal{R}(x) \to L(x) = N^2/\mathcal{R}(x)$ (linear), or λ(i, x) (nonlinear).
2. Pick energy (if λ or Φ is given) or coenergy (if i or MMF is given).
3. Differentiate with respect to x (or θ), holding the right variable constant; only then substitute numbers.
4. Interpret the sign: positive f_{fld} acts toward increasing x.

Mistakes: differentiating $\frac{1}{2}Li^2$ at constant λ (wrong sign); substituting numbers before differentiating; using $W = \frac{1}{2}\lambda i$ for a nonlinear coil; forgetting that the force is on the *moving part* and its sign convention; adding losses into W_f.

7. Worked examples

Example 22.1 — Force on a conductor

Given. A 0.3 m conductor carries 50 A along $-\hat{z}$ in $B = 0.8\hat{y}$ T. **Find** the force. **Solution.** $\mathbf{F} = 50(-0.3\hat{z}) \times (0.8\hat{y}) = +12\hat{x}$ N. $|F| = BIl = 12$ N.

Example 22.2 — Direction of rotation (LN12 slide 5)

Given. Figure 22.1 with $B_0 = 0.5$ T, $I = 10$ A, $l = 0.2$ m, $R = 0.05$ m. **Find** the torque at α = 30°, 90°, 150°. **Solution.** From (22.2): $T = -2(0.5)(10)(0.2)(0.05)\sin\alpha = -0.1\sin\alpha$ N·m: -0.05, -0.10, -0.05 N·m (clockwise). The rotor turns clockwise until α = 0 and stays there (or oscillates about it if undamped). To keep rotating, the currents must reverse when the conductors cross the α = 0 line — commutation (LN15 slides 11, 20).

Example 22.3 — Energy and coenergy give the same force (Sen Example 3.2)

Given. A nonlinear system with $i = (\lambda g/0.09)^2$, valid for $0 < i < 4$ A and $3 < g < 10$ cm. **Find** the force on the moving part at $i = 3$ A, $g = 5$ cm, by coenergy and by energy. **Solution.** $\lambda = 0.09\sqrt{i}/g$.

Coenergy: $W'_f = \int_0^i \lambda\, di = \dfrac{0.09}{g}\cdot\dfrac{2}{3}i^{3/2} \Rightarrow f = \partial W'_f/\partial g|_i = -\dfrac{0.06\, i^{3/2}}{g^2} = -\dfrac{0.06(5.196)}{0.0025} = -124.7$ N.

Energy: $W_f = \int_0^\lambda (\lambda g/0.09)^2 d\lambda = \dfrac{g^2\lambda^3}{3(0.09)^2} \Rightarrow f = -\partial W_f/\partial g|_\lambda = -\dfrac{2g\lambda^3}{3(0.0081)}$ with $\lambda = 3.118$ Wb·t: -124.7 N. **Interpretation.** Negative: the force acts to decrease g (close the gap). Check: $W_f = 3.12$ J, $W'_f = 6.24$ J, sum $= \lambda i = 9.35$ J ✓; energy ≠ coenergy because the system is nonlinear.

Example 22.4 — Energy bookkeeping for slow and fast motion (original)

Given. A linear actuator with $L(x) = 0.2/(1 + x/0.002)$ H moves from $x_1 = 2$ mm to $x_2 = 1$ mm (gap closing). (a) At constant 5 A: find electrical input, change in stored energy and mechanical work. (b) At constant λ (starting at 5 A at x_1): the same. **Solution.** $L(x_1) = 0.1$ H, $L(x_2) = 0.1333$ H. (a) $\Delta\lambda = 5(0.0333) = 0.1667$ Wb·t; electrical input $i\Delta\lambda = 0.833$ J; $\Delta W_f = \frac{1}{2}(0.0333)(25) = 0.417$ J; mechanical work 0.417 J — exactly half. (b) $\lambda = 0.5$ Wb·t constant: $\Delta W_f = \lambda^2/(2L_2) - \lambda^2/(2L_1) = 0.9375 - 1.25 = -0.3125$ J; electrical input 0; mechanical work +0.3125 J (and the current falls to 3.75 A). **Interpretation.** The large displacement makes the two works differ; for a vanishingly small displacement they agree (the script checks a 0.1 µm step to 10^{-4}), which is why (22.7) defines one force.

Example 22.5 — Field energy of an actuator with a nonlinear core (Sen Example 3.1)

Given. Actuator of Sen Fig. E3.1: cast steel, 250 turns, coil resistance 5 Ω, two gaps of 5 mm, core path 60 cm, steel volume 0.003 m³, gap volume $2(0.05 \times 0.10 \times 0.005)$ m³; $B = 1.0$ T in the gaps. **Find** the DC source voltage and the stored field energy. **Solution.** Cast steel at 1.0 T: $H_c = 670$ A/m. $Ni = 670(0.6) + (1.0/\mu_0)(0.01) = 402 + 7958 = 8360$ At ⇒ $i = 33.44$ A ⇒ $V = 167.2$ V. Energy: core $\approx \frac{1}{2}(1.0)(670) \times 0.003 = 1.0$ J (Sen approximates $\int H\, dB$ by a triangle); gaps $\dfrac{1.0^2}{2\mu_0} \times 5 \times 10^{-5} = 19.9$ J; total 20.9 J — 95 % in the air gaps.

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8. Optional practice

P22.1. A conductor 0.5 m long moves at 10 m/s perpendicular to a 0.6 T field and carries 20 A. Find the motional EMF, the force, and show $Fv = eI$.

▼ **Solution**

$e = Blv = 3$ V; $F = BIl = 6$ N; $Fv = 60$ W $= eI = 60$ W. Whether this is motoring or generating depends on the current direction relative to e.

P22.2. For $\lambda = 0.5i/(1 + 0.1x)$ (x in mm), find the force at $i = 4$ A, $x = 2$ mm, using coenergy.

▼ **Solution**
Linear: $L(x) = 0.5/(1 + 0.1x)$ H with x in mm ⇒ $dL/dx = -0.05/(1 + 0.1x)^2$ H/mm $= -50/(1.2)^2 = -34.7$ H/m. $f = \frac{1}{2}(16)(-34.7) = -278$ N (acts to decrease x).

P22.3 (concept). Why can the energy method give the force on an iron bar that carries no current, whereas $\mathbf{J} \times \mathbf{B}$ with only the coil current cannot?

▼ **Solution**
The bar's magnetization changes the reluctance and hence the stored energy/coenergy as it moves; the energy method captures any force that changes the stored energy. A Lorentz-force calculation would have to include the equivalent magnetization currents in the iron (or use the Maxwell stress), which the simple BIl formula ignores.

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9. Summary

Relation	Validity
$\mathbf{F} = I\mathbf{l} \times \mathbf{B}$; $e = Blv$	straight conductor, uniform B
$dW_f = i\,d\lambda - f\,dx$	lossless coupling field
$W_f = \int_0^\lambda i\,d\lambda$, $W'_f = \int_0^i \lambda\,di$, $W_f + W'_f = \lambda i$	fixed position
$f = -\partial W_f/\partial x\|_\lambda = +\partial W'_f/\partial x\|_i$	general
$f = \frac{1}{2} i^2\, dL/dx$	linear
Slow motion: work = ΔW′; fast motion: work = −ΔW_f	interpretation

10. Vocabulary

English	Türkçe
electromechanical energy conversion	elektromekanik enerji dönüşümü
Lorentz force	Lorentz kuvveti
reluctance force	relüktans kuvveti
motional (speed) voltage	hareket (hız) gerilimi
energy balance	enerji dengesi
coupling field	kuplaj alanı
field energy; coenergy	alan enerjisi; koenerji
actuator	eyleyici (aktüatör)
right-hand rule	sağ el kuralı

11. Sources

- LN12 viewer pp. 4–31 (slides 2–15): Lorentz force and applications, direction of rotation, reluctance-force question, basics, losses, lossless system, energy balance, energy/coenergy. Notes.
- LN15 viewer pp. 11–14 (slides 8–10): rotor in uniform field, macroscopic Lorentz force.
- Sen §3.1–3.3 pp. 93–103 (viewer 117–127): Eqs. 3.1–3.34, Examples 3.1–3.2, Figs. 3.4–3.5; §4.1 pp. 119–121: Eqs. 4.1–4.2.
- Verification: `lesson22_energy_coenergy.py`.

Lesson 23 — Force calculation in linear electromagnetic systems; magnetic stress

Chapter IV · Week 10 · Lesson 23 of 31

Assistant-written teaching note. Teacher sources: LN12 slides 17–26 and 36 (force from stored energy, plunger example, junkyard magnet, virtual work at constant Φ and constant MMF, summary); LN13 slides 2–4 (magnetic stress). Book: Sen §3.3–3.3.1 (printed pp. 99–107, viewer 123–131), Examples 3.3–3.4. Exams: MT I Fall 2016 Q1 (crane magnet); Jan 2022 Final Q2 (EV contactor); LN12 plunger exercise. Numbers: `lesson23_force_linear.py`.

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1. Place and goals

Lesson 22 established $f = \frac{1}{2} i^2 \, dL/dx$ for linear systems. Here we turn it into a routine for actuators — relays, solenoids, lifting magnets, contactors, loudspeakers — and add the quick **magnetic stress** formula $B^2/(2\mu_0)$. Linear means the iron is either infinitely permeable or operated below saturation, so $L(x)$ does not depend on current.

You should be able to:

1. Express $\mathcal{R}(x)$ and $L(x)$ of an actuator from its geometry and differentiate them.
2. Compute force by virtual work at constant MMF (coenergy), at constant flux (energy) and by magnetic stress, and show they agree.
3. Handle AC excitation (average force) and combine magnetic force with springs, gravity and inertia.
4. Solve multi-part exam problems such as the EV contactor and the crane electromagnet.

Prerequisites: Lessons 12–14 (reluctance circuits, inductance), Lesson 22.

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2. The method of virtual work in practice

Step 1 — magnetic circuit as a function of position. Write each reluctance in terms of the moving coordinate x. Typical cases:

- *Gap length varies* (relay armature, crane magnet, contactor): $\mathcal{R}_g = x/(\mu_0 A)$, so $\mathcal{R}$ grows linearly with x.
- *Overlap area varies* (plunger moving sideways, LN12 slides 18–19): $\mathcal{R}_g = g/(\mu_0 l(d-x))$.
- *Angle varies* (rotating systems, Lesson 24).

Step 2 — inductance. $L(x) = N^2/\mathcal{R}_{tot}(x)$.

Step 3 — force.

$$f = \frac{1}{2} i^2 \frac{dL}{dx} \ \text{(constant } i\text{)} \qquad = -\frac{1}{2}\Phi^2 \frac{d\mathcal{R}}{dx} \ \text{(constant } \Phi\text{)} \qquad \text{(LN12 slides 22, 25–26; LN14 slide 2)} \tag{23.1}$$

The two are identical at a given operating point: with $\Phi = Ni/\mathcal{R}$, $\frac{1}{2} i^2 \frac{d}{dx}\left(\frac{N^2}{\mathcal{R}}\right) = -\frac{1}{2}\frac{N^2 i^2}{\mathcal{R}^2}\frac{d\mathcal{R}}{dx} = -\frac{1}{2}\Phi^2 \frac{d\mathcal{R}}{dx}$.

LN12's summary (slide 36) gives the physical rule: **the force always acts to increase inductance / decrease reluctance.** If a displacement would shorten a gap or enlarge an overlap, the force pushes that way.

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3. Magnetic stress (LN13 slides 3–4)

Consider a gap of length x and area A carrying a uniform flux density B, with the iron assumed infinitely permeable. At constant flux the stored energy in the gap is $W_f = \dfrac{B^2}{2\mu_0}Ax$, so

$$f = -\left.\frac{\partial W_f}{\partial x}\right|_{\Phi} = -\frac{B^2}{2\mu_0}A \quad\Longrightarrow\quad P = \frac{B^2}{2\mu_0} = \frac{\mu_0 H^2}{2}\ \ [\mathrm{N/m^2}], \qquad F = \frac{B^2 A}{2\mu_0} \tag{23.2}$$

(Sen Eq. 3.38: "magnetic pressure"). The minus sign means the force pulls the faces together (reduces x). The stress is **independent of the area** (LN13 slide 3): each square metre of pole face pulls with $B^2/(2\mu_0)$ — 398 kPa (about 4 atmospheres) at 1 T, 1.02 MPa at 1.6 T. With several gaps, add the force of each face. This is the quickest exam route whenever the gap flux density is known, and it explains LN13 slide 4: a steel backplate on a magnet closes the flux path behind it, raising the flux density at the working face.

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4. AC excitation

The force depends on B^2 (or i^2), so it is always attractive and pulsates at **twice** the supply frequency. Its average uses the RMS value: $\bar{F} = \dfrac{B_{rms}^2 A}{2\mu_0}$ (Sen Eq. 3.38c). On an AC supply the coil current is limited by the inductive reactance, not just the resistance, so the average force can be much smaller than on a DC supply of the same voltage (Example 23.4). Lifting magnets and contactors with heavy duty are therefore DC-operated; AC contactors use shading rings to avoid chatter at the force zeros.

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5. Combining with mechanics

The magnetic force joins the other forces on the moving mass (LN14 slides 19–22):

$$m\frac{d^2x}{dt^2} = f_{mag}(x, i) + f_{spring} + f_{gravity} - B_d\frac{dx}{dt}. \tag{23.3}$$

Static questions ("minimum current to hold", "will it close?") set the left side to zero; dynamic questions ("initial acceleration", "closing time") use (23.3) directly. Because $f_{mag} \propto 1/x^2$ for a gap-dominated actuator at constant current, the magnetic force grows rapidly as the gap closes, while a spring force grows only linearly: once the armature starts moving, it usually snaps shut (Figure 23.3).

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6. Common mistakes

- Forgetting there are two (or three) gaps in series or in parallel.
- Using A of one gap when the flux crosses several faces of different area (use stress per face, or total inductance).
- Mixing signs: compute magnitude and decide the direction from "increase L".

- Applying the linear formula when the core saturates (exam problems usually instruct to stay in the linear region and ask for the corresponding current limit).
- For AC, using the peak current with the RMS formula or forgetting the coil reactance.

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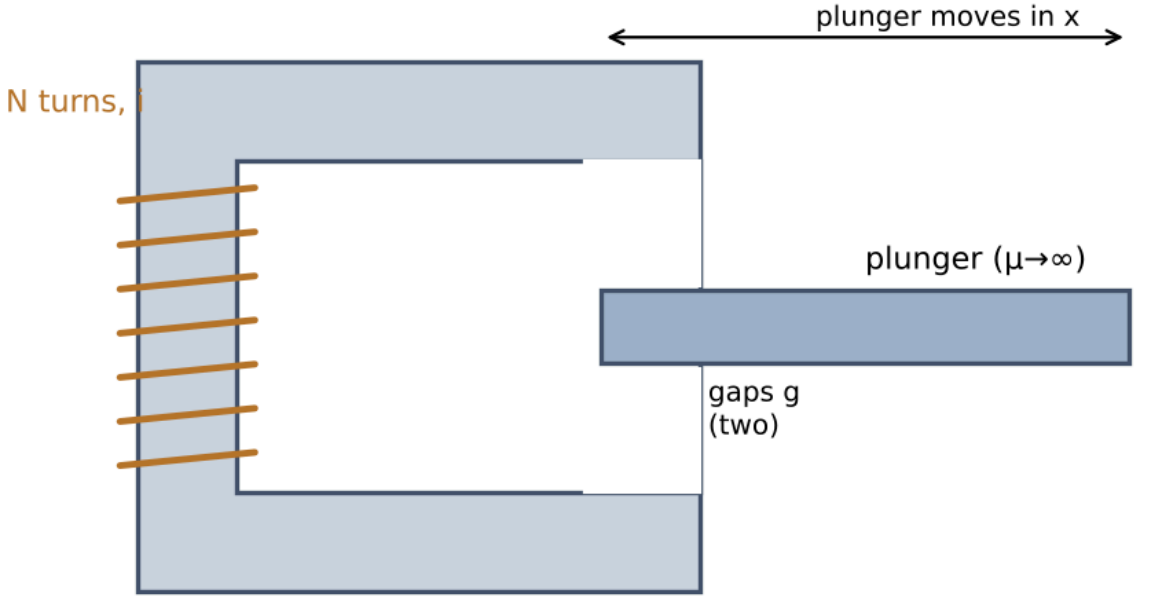
7. Worked examples

Example 23.1 — Magnetic pressure

Find the pressure at 0.5, 1.0 and 1.6 T. **Solution.** $B^2/(2\mu_0)$ = 99.5 kPa, 397.9 kPa, 1.019 MPa. A 1.6 T pole face of 10 cm² pulls with about 1 kN.

Example 23.2 — The LN12 plunger: energy and force versus position (LN12 slides 18–19, 23)

Given. Figure 23.1: lossless $N = 1000$-turn coil on a core with μ → ∞; plunger (μ → ∞) of width $d = 15$ cm, depth $l = 10$ cm; two air gaps of length $g = 2$ mm between the pole faces and the plunger; overlap $(d - x)$; $I = 10$ A. **Find** the stored energy as a function of x, and the force.



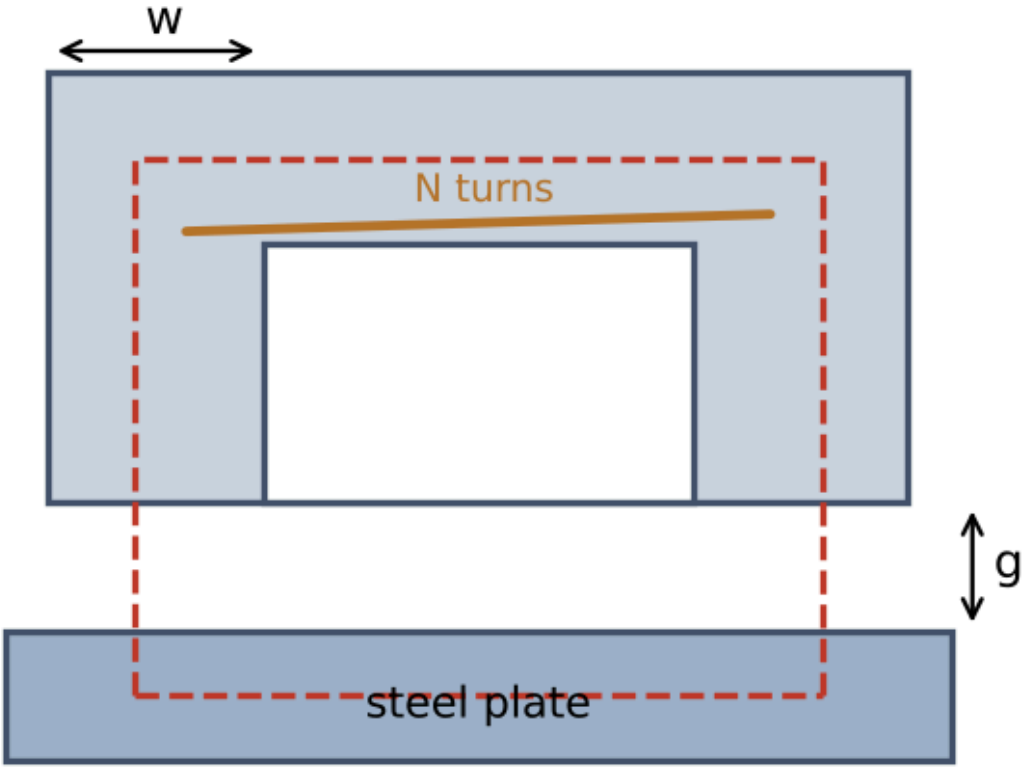
Solution. Two gaps in series, each of area $l(d - x)$: $\mathcal{R}(x) = \dfrac{2g}{\mu_0 l(d-x)}$, so

$$L(x) = \frac{\mu_0 N^2 l(d-x)}{2g} = 31.4\,(d-x)\ \text{H} \quad (4.712\ \text{H at } x = 0).$$

$W_f(x) = \tfrac{1}{2}L(x)I^2 = 235.6\,(1 - x/d)$ J — falling linearly as the plunger is withdrawn. Force: $f = \tfrac{1}{2}I^2\dfrac{dL}{dx} = -\dfrac{\mu_0 N^2 l I^2}{4g} = -1571$ N, independent of x while the overlap is not zero. **Check by stress.** $B_g = \mu_0 NI/(2g) = 3.14$ T (a deliberately idealized number — real iron would saturate near 2 T); the force along x equals the energy density times the rate at which gap volume is lost: $\dfrac{B_g^2}{2\mu_0} \times (2g\,l) = 1571$ N ✓. **Interpretation.** Negative: the plunger is pulled *in* (toward larger overlap, larger L). This sideways "pull-in" force is how solenoid actuators and loudspeaker-like movers work (LN12 slides 27–28).

Example 23.3 — Crane (junkyard) magnet (LN12 slide 24; MT I Fall 2016 Q1)

Given. Figure 23.2: C-core with N turns carrying I; iron piece below separated by two gaps of length g, each of area $A = wD$; iron μ → ∞. **Derive** the force, then evaluate for the MT 2016 numbers: $N = 500$, $I = 20$ A, $g = 0.5$ cm, $A = 100$ cm^2.



two gaps of area A = wD: F = μ0N²I²A/(4g²) = 2·B²A/(2μ0)

Solution. $\mathcal{R} = 2g/(\mu_0 A)$, $L(g) = \mu_0 N^2 A/(2g)$, so

$$f = \frac{1}{2} I^2 \frac{dL}{dg} = -\frac{\mu_0 N^2 I^2 A}{4g^2}.$$

Numbers: $f = -\dfrac{4\pi \times 10^{-7}(500)^2(20)^2(0.01)}{4(0.005)^2} = -12\,566$ N — enough to hold about 1.28 t. Stress check: $B = \mu_0 NI/(2g) = 1.257$ T; two faces: $2 \times \dfrac{B^2 A}{2\mu_0} = 12\,566$ N ✓. **Constant current vs constant flux.** If the gap halves at constant current, B doubles and the force quadruples (50 kN on paper; in reality the iron saturates). At constant flux the force is unchanged, because the stress depends only on B.

Example 23.4 — Force between two pole faces (Sen Example 3.3)

Given. $N = 500$, $i = 2$ A, gap 2 cm × 2 cm × 1 mm, core reluctance, leakage and fringing neglected. **Find** the force and the stored energy. **Solution.** $B_g = \mu_0 Ni/l_g = 1.257$ T; $F = \dfrac{B_g^2}{2\mu_0} A = \dfrac{\mu_0 N^2 i^2 A}{2 l_g^2} = 251.3$ N; $W_f = \dfrac{B_g^2}{2\mu_0} A l_g = 0.251$ J.

Example 23.5 — Lifting magnet on DC and on AC (Sen Example 3.4)

Given. Two gaps, each 6 cm × 6 cm, held at 5 mm; 300 turns, 6 Ω; core reluctance and fringing neglected. (a) 120 V DC: stored energy and lifting force. (b) 120 V rms, 60 Hz: average force.

Solution. (a) $i = 120/6 = 20$ A; $B_g = \mu_0 N i/(2g) = 0.754$ T; $W_f = \dfrac{B_g^2}{2\mu_0}(2 \times 36 \times 10^{-4} \times 5 \times 10^{-3}) = 8.14$ J; $F = \dfrac{B_g^2}{2\mu_0}(2 \times 36 \times 10^{-4}) = 1629$ N.

b. $L = N^2\mu_0 A/(2g) = 40.7$ mH; $Z = 6 + j15.35\ \Omega$; $I_{rms} = 120/16.48 = 7.28$ A; $B_{rms} = 0.2745$ T; $\bar{F} = \dfrac{B_{rms}^2}{2\mu_0}(2A) = 216$ N (Sen 216.3 N with rounded intermediate values) — about one-eighth of the DC force. **Check.** Time-averaging $B(t)^2/(2\mu_0) \cdot 2A$ over a period reproduces the RMS formula. **Interpretation.** On AC the coil's reactance (15.3 Ω) limits the current; the force also pulses at 120 Hz.

Example 23.6 — EV battery contactor (Jan 2022 Final Q2)

Given. (exams/archive/Final/EE361_2022_Final_Exam_Q4_solution.pdf pp. 6–8.) E-shaped stationary core with a 200-turn coil on the centre leg and a movable E-shaped armature above it; gap y (same at all three legs) varies from 5 mm (open) to 1 mm (contacts closed). Mean lengths $l_A = 5.5$, $l_B = 7.5$, $l_C = 2.5$, $l_D = 4.5$ cm; side legs $w_s = 1$ cm, centre leg $w_m = 2$ cm, axial length 1 cm (side-leg area 1 cm², centre 2 cm²). Core B–H linear to $B_{sat} = 1.6$ T at $H_{sat} = 160$ A/m ⇒ $\mu = 0.01$ H/m. Moving mass 0.125 kg; two springs, each $F = 2500(7 \times 10^{-3} - y)$ N upward; $g = 9.8$ m/s². Neglect leakage, fringing, friction. **Find** (a) the equivalent circuit and reluctances; (b) $\mathcal{R}_{eq}(y)$; (c) the maximum current at $y = 1$ mm without saturation; (d) the initial acceleration with 10 A at $y = 5$ mm; (e) the minimum current that holds the contacts closed at $y = 1$ mm.

Solution. (a) MMF Ni in the centre leg; centre branch: $\mathcal{R}_{mid} = \dfrac{l_C + l_D}{\mu A_m} + \dfrac{y}{\mu_0 A_m}$; each side branch: $\mathcal{R}_{side} = \dfrac{l_A + l_B}{\mu A_s} + \dfrac{y}{\mu_0 A_s}$; the two side branches in parallel, in series with the centre branch.

b. $\mathcal{R}_{eq} = \mathcal{R}_{mid} + \mathcal{R}_{side}/2$. Because $A_m = 2A_s$ and half the flux returns through each side, this simplifies to

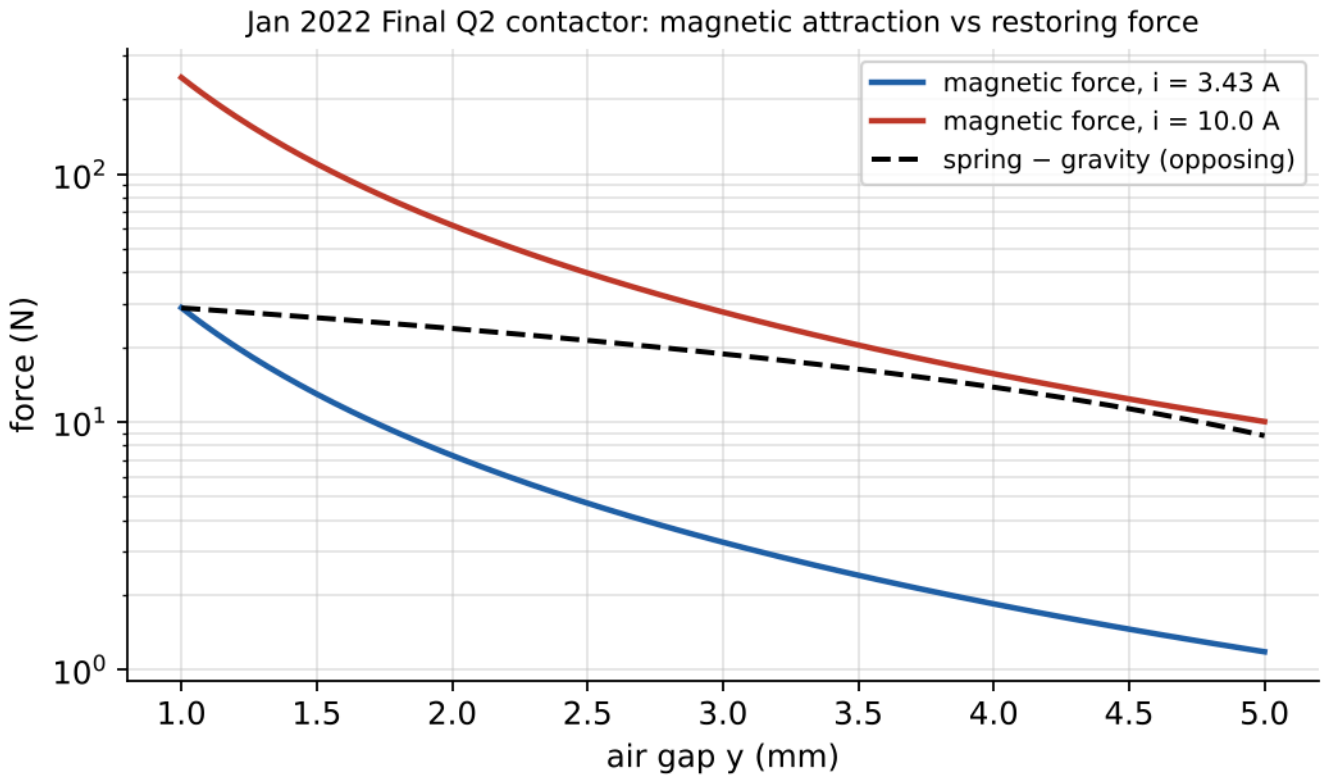
$$\mathcal{R}_{eq}(y) = \frac{l_A + l_B + l_C + l_D}{\mu(2 \times 10^{-4})} + \frac{y}{\mu_0(1 \times 10^{-4})} = 1.0 \times 10^5 + 7.958 \times 10^9\, y \ \ \text{A/Wb}.$$

At y = 1 mm: 8.06×10^6 A/Wb; at 5 mm: 3.99×10^7 A/Wb. The flux through $\mathcal{R}_{eq}$ is the centre-leg flux; each side leg carries half.

c. Saturation first appears where B is highest; all legs have the same B (half flux in half area), so require $B = 1.6$ T in the centre leg: $\Phi = 1.6 \times 2 \times 10^{-4} = 3.2 \times 10^{-4}$ Wb ⇒ $i_{max} = \Phi\mathcal{R}_{eq}/N = 12.9$ A.

d. $f = \frac{1}{2}i^2\dfrac{dL}{dy} = -\frac{1}{2}i^2N^2\dfrac{\mathcal{R}'}{\mathcal{R}^2}$ with $\mathcal{R}' = 7.958 \times 10^9$: at 10 A, 5 mm, $|f| = 10.0$ N toward smaller y (stress check: $B = 0.251$ T in all three gaps, $\dfrac{B^2}{2\mu_0}(2+1+1) \times 10^{-4} = 10.0$ N ✓). Springs: $2 \times 2500 \times 0.002 = 10.0$ N upward; gravity 1.225 N downward. Net 1.228 N downward ⇒ **initial acceleration 9.8 m/s²** toward closing.

e. At y = 1 mm the springs push $2 \times 2500 \times 0.006 = 30$ N up; gravity helps with 1.225 N: the magnet must supply 28.78 N: $i^2 = \dfrac{2f\mathcal{R}^2}{N^2\mathcal{R}'}$ ⇒ $i_{min} = 3.43$ A (below the 12.9 A saturation limit, so the linear model is valid). **Extension (dynamics).** Integrating (23.3) at a constant 10 A from rest at 5 mm, the armature closes in 23 ms and hits the stop at 1.4 m/s. Figure 23.3 shows why: the magnetic force rises as $1/\mathcal{R}^2$ while the spring force rises only linearly.



8. Optional practice

P23.1. An electromagnet must hold a 50 kg steel plate across two faces of 5 cm × 5 cm with a 0.2 mm residual gap at each face. What flux density and what MMF are needed (iron μ → ∞)?

▼ **Solution**
Required force 490.5 N over $2 \times 25 \times 10^{-4}$ m²: $B^2/(2\mu_0) = 98\,100$ Pa ⇒ $B = 0.497$ T. MMF $= 2gB/\mu_0 = 2(0.0002)(0.497)/\mu_0 = 158$ At.

P23.2. In Example 23.6, what current is needed to *start* closing the contactor from $y = 5$ mm (zero net force)?

▼ **Solution**
Need $f_{mag} = 10 - 1.225 = 8.775$ N at 5 mm: $i = 10\sqrt{8.775/10.003} = 9.37$ A.

P23.3 (concept). Why does a DC relay draw the same current open or closed, while an AC relay draws much less current once closed?

▼ **Solution**
DC: current $= V/R$, independent of the gap. AC: current $\approx V/(\omega L)$; closing the gap raises L greatly, so the current drops — AC coils must be designed so they do not overheat if the armature is blocked open.

9. Summary

Relation	Validity
$f = \frac{1}{2}i^2\,dL/dx = -\frac{1}{2}\Phi^2\,d\mathcal{R}/dx$	linear
$P = B^2/(2\mu_0)$, $F = B^2A/(2\mu_0)$ per face	iron μ → ∞, uniform gap field

Relation	Validity
Gap actuator, two gaps: $F = \mu_0 N^2 I^2 A/(4g^2)$	μ → ∞
AC: $\bar{F} = B_{rms}^2 A/(2\mu_0)$, pulsates at $2f$	sinusoidal
$m\ddot{x} = f_{mag} + f_{spring} + f_g - B_d\dot{x}$	dynamics

10. Vocabulary

English	Türkçe
virtual work	sanal iş
magnetic stress (pressure)	manyetik gerilme (basınç)
plunger; armature (of a relay)	dalgıç (piston); armatür
solenoid; relay; contactor	solenoid; röle; kontaktör
lifting magnet	kaldırma mıknatısı
spring constant	yay sabiti
holding current	tutma akımı

11. Sources

- LN12 viewer pp. 32–50 (slides 16–26), p. 66 (slide 36): singly excited system, plunger example (energy vs x, force vs x), force formulas, junkyard magnet, virtual work at constant Φ and MMF. LN13 viewer pp. 2–5 (slides 2–4): magnetic stress. Notes, LN13.
- Sen §3.3.1 pp. 102–107 (viewer 126–131): Eqs. 3.28–3.38c, Examples 3.3–3.4.
- MT I Fall 2016 Q1 (exams/archive/MT/MT I - Fall2016.pdf pp. 2–4); Jan 2022 Final Q2 (pp. 6–8).
- Verification: `lesson23_force_linear.py` (includes an ODE simulation of contactor closing).

Lesson 24 — Torque in rotating systems, reluctance motors, multiply excited systems and dynamic equations

Chapter IV · Week 10 · Lesson 24 of 31

Assistant-written teaching note. Teacher sources: LN13 slides 5–20 (mechanical power and energy, rotational inertia, singly excited rotating system, torque from energy/coenergy, reluctance motors, summary); LN14 (all 32 slides: multiply excited systems, coupled energy, torque components, cylindrical/salient configurations, mechanical dynamic equations). Book: Sen §3.4–3.5 (printed pp. 108–113, viewer 132–137), Example 3.5. Exams: Final Fall 2014 Q2; Final Fall 2015 Q2 (Dyson motor); Final Fall 2018 Q1. Note: LN14's geometry drawings (slides 4–9, 15–17) are missing in the supplied print; the configurations are redrawn from Sen Figs. 3.6–3.7 and the inductance functions given in the exams. Numbers: `lesson24_torque_multiply_excited.py`.

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1. Place and goals

Most converters rotate. Fortunately rotation needs no new physics: "rotational systems are similar to linear systems, but take the derivative with respect to θ instead of x" (LN13 slide 20). This lesson applies the energy method to rotors, explains the reluctance motor and the doubly excited machine whose torque terms reappear in every AC machine (EE362), and couples the electromagnetic torque to the mechanical equation of motion.

You should be able to:

1. Translate linear mechanics to rotation (θ, ω, T, J) and compute mechanical power and energy.
2. Derive the torque of singly excited systems, $T = \frac{1}{2} i^2 \, dL/d\theta$, and apply it to a reluctance rotor with DC and AC excitation.
3. Derive the coenergy and torque of multiply excited systems, identify reluctance and mutual (alignment) torque components, and specialize to cylindrical/salient rotors and stators.
4. Find the speeds at which an AC-excited machine produces average torque.
5. Write and solve the mechanical dynamic equation with inertia, damping, spring and load.

Prerequisites: Lessons 14 (coupled energy), 22–23.

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2. Rotational mechanics (LN13 slides 5–6; LN14 slide 18)

Linear	Rotational
position x (m)	angle θ (rad)
velocity $v = dx/dt$ (m/s)	angular velocity ω = dθ/dt (rad/s)
force F (N)	torque T (N·m)
mass m (kg)	moment of inertia J (kg·m²)
$F = m\,dv/dt$	$T = J\,d\omega/dt$
power $P = Fv$	$P = T\omega$

Linear	Rotational
work $W = \int F\,dx$	$W = \int T\,d\theta$
kinetic energy $\frac{1}{2}mv^2$	$\frac{1}{2}J\omega^2$

(LN13 slide 6 labels the torque equation with the unit “Watt” — the unit of torque is N·m; errata E4.) Speed in rpm: $\omega = 2\pi n/60$.

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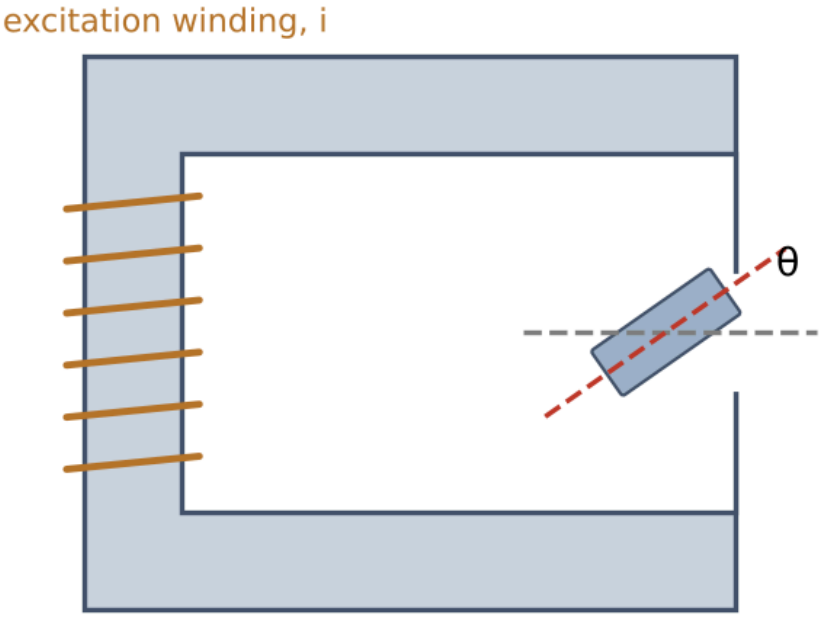
3. Singly excited rotating systems

3.1 Torque from energy and coenergy

A stator C-core with an excitation winding and a rotor of soft iron between its poles (LN13 slides 7–8, Figure 24.1). The rotor angle θ is measured between the stator and rotor magnetic axes. The energy balance becomes $dW_f = i\,d\lambda - T\,d\theta$, so (LN13 slides 9–11):

$$T = -\left.\frac{\partial W_f(\lambda,\theta)}{\partial\theta}\right|_{\lambda} = +\left.\frac{\partial W'_f(i,\theta)}{\partial\theta}\right|_{i}, \qquad \text{linear: } T = \frac{1}{2}i^2\frac{dL(\theta)}{d\theta} = -\frac{1}{2}\Phi^2\frac{d\mathcal{R}(\theta)}{d\theta}. \tag{24.1}$$

(LN13 slide 10 writes “f =” for the coenergy form; it is the torque — errata E5.)



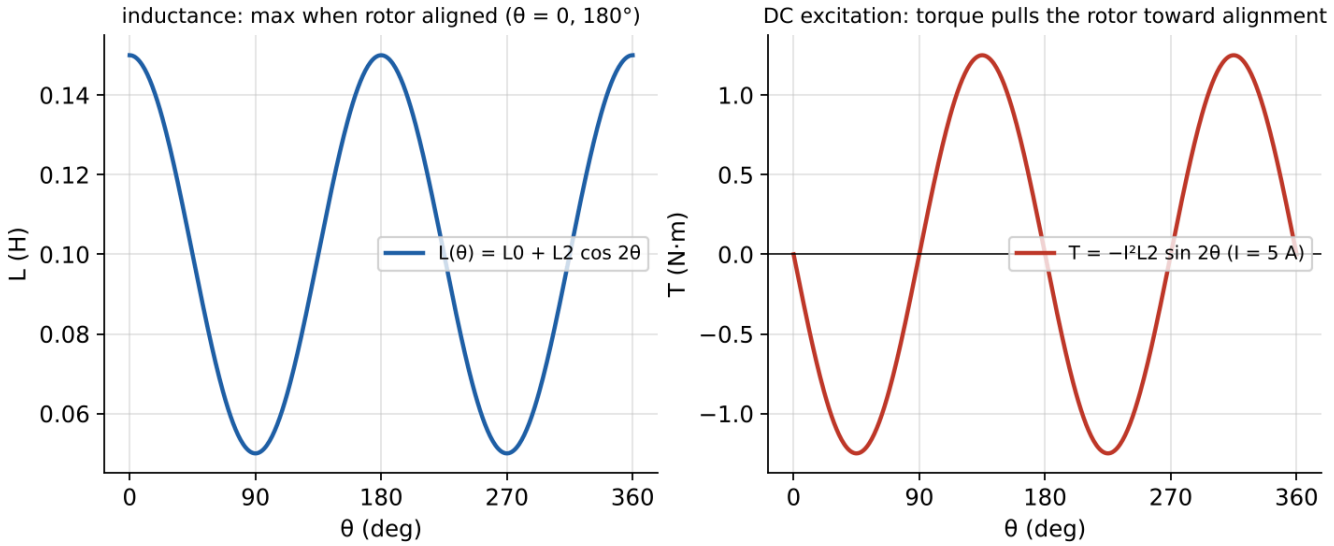
stator axis (dashed grey) and rotor axis (red); torque turns the rotor toward θ = 0

3.2 The reluctance rotor

Its inductance is largest when the rotor is aligned with the stator poles (θ = 0 or π, small gap, low reluctance) and smallest in quadrature (θ = ±π/2). A good model, periodic with period π, is

$$L(\theta) = L_0 + L_2\cos 2\theta, \qquad L_d = L_0 + L_2,\ L_q = L_0 - L_2 \qquad \text{(Sen Ex. 3.5)}. \tag{24.2}$$

DC current I: $T = \frac{1}{2}I^2(-2L_2\sin 2\theta) = -I^2L_2\sin 2\theta$. The torque pulls the rotor toward the nearest aligned position (θ = 0 or π) — “what would happen in this circuit?” (LN13 slide 7): the rotor turns to align and stays there. No continuous rotation is possible with DC (Figure 24.2).



Continuous rotation (LN13 slides 12–19). Either switch the current on only during the intervals when $dL/d\theta > 0$ in the direction of motion (switched-reluctance motors, Dyson's "digital motor", stepper motors with several stator coils energized in sequence), or feed a sinusoidal current and rotate at synchronous speed:

Sinusoidal current $i = I_m \sin \omega t$ with $\theta = \omega_m t + \delta$:

$$T = -I_m^2 L_2 \sin 2(\omega_m t + \delta) \sin^2 \omega t = -\frac{I_m^2 L_2}{2}\Big[\sin 2(\omega_m t + \delta) - \tfrac{1}{2}\sin 2\big((\omega_m + \omega)t + \delta\big) - \tfrac{1}{2}\sin 2\big((\omega_m - \omega)t + \delta\big)$$

Each sinusoid averages to zero unless its frequency is zero. So average torque exists only at $\omega_m = 0$ (holding torque $-\frac{1}{2} I_m^2 L_2 \sin 2\delta$) and at **synchronous speed** $\omega_m = \pm\omega$, where $T_{avg} = \frac{1}{4} I_m^2 L_2 \sin 2\delta = \frac{1}{8} I_m^2 (L_d - L_q) \sin 2\delta$ (Sen Ex. 3.5). The machine is a single-phase synchronous reluctance motor: it must be brought to speed, then runs synchronously; δ (the "torque angle") adjusts to the load. This answers LN13 slide 19 ("calculate the torque expression with constant current / with sinusoidal current").

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4. Multiply excited systems (LN14)

4.1 Energy and coenergy with two windings

With a stator winding (1) and a rotor winding (2), both linear (LN14 slides 9–13):

$$dW_f = i_1 d\lambda_1 + i_2 d\lambda_2 - T\, d\theta, \qquad \lambda_1 = L_{11} i_1 + M i_2, \;\; \lambda_2 = M i_1 + L_{22} i_2,$$
$$W_f' = W_f = \tfrac{1}{2} L_{11}(\theta) i_1^2 + \tfrac{1}{2} L_{22}(\theta) i_2^2 + M(\theta) i_1 i_2 = \tfrac{1}{2} \mathbf{I}^T \mathbf{L}(\theta) \mathbf{I}. \tag{24.3}$$

LN14 slide 11 stresses that the energy "in coil 1" is *not* $\frac{1}{2} L_{11} i_1^2$ when the other coil carries current; only the total (24.3) is meaningful.

4.2 Torque

$$\boxed{T = \left.\frac{\partial W_f'}{\partial \theta}\right|_{i_1, i_2} = \underbrace{\frac{1}{2} i_1^2 \frac{dL_{11}}{d\theta} + \frac{1}{2} i_2^2 \frac{dL_{22}}{d\theta}}_{\text{reluctance torque}} + \underbrace{i_1 i_2 \frac{dM}{d\theta}}_{\text{mutual (alignment) torque}} = \frac{1}{2} \mathbf{I}^T \frac{d\mathbf{L}}{d\theta} \mathbf{I}} \tag{24.4}$$

(LN14 slide 14; Sen Eq. 3.44.) The first two terms exist only if a self-inductance varies with θ (saliency); the third exists whenever the mutual inductance varies — the two windings try to align their magnetic axes.

4.3 Special configurations (LN14 slides 15–17)

Configuration	Varying inductances	Torque
cylindrical rotor, cylindrical stator (uniform gap)	only M	$T = i_1 i_2\, dM/d\theta$
cylindrical rotor, salient stator	L_{22} (rotor coil sees varying gap) and M	$T = \frac{1}{2} i_2^2\, dL_{22}/d\theta + i_1 i_2\, dM/d\theta$
salient rotor, cylindrical stator	L_{11} and M	$T = \frac{1}{2} i_1^2\, dL_{11}/d\theta + i_1 i_2\, dM/d\theta$
salient rotor, no rotor winding	L_{11}	reluctance motor, Section 3.2

With a uniform air gap and $M = M_{max}\cos\theta$ (Sen Eq. 3.46), $T = -i_1 i_2 M_{max}\sin\theta$: two DC currents pull the axes into alignment (θ = 0) — like two bar magnets.

4.4 Conditions for average torque with AC currents (Sen §3.5)

Let $i_s = I_{sm}\cos\omega_s t$, $i_r = I_{rm}\cos(\omega_r t + \alpha)$, $\theta = \omega_m t + \delta$, $M = M_{max}\cos\theta$. Expanding $T = -I_{sm} I_{rm} M_{max}\cos\omega_s t\cos(\omega_r t + \alpha)\sin(\omega_m t + \delta)$ gives four sinusoids of frequencies $\omega_m \pm (\omega_s \pm \omega_r)$; an average exists only when one frequency is zero:

$$|\omega_m| = |\omega_s \pm \omega_r| \qquad \text{(Sen Eq. 3.52)}.$$

- $\omega_r = 0$ (DC rotor), $\omega_m = \pm\omega_s$: the **synchronous machine**; $T_{avg} = -\frac{1}{2} I_{sm} I_R M_{max}\sin\delta$; no average torque at standstill (not self-starting); single-phase torque pulsates at $2\omega_s$ — why large machines are polyphase.
- $\omega_m = \omega_s - \omega_r$ with AC on both: the principle of the **induction machine** (EE362).

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5. Dynamic equations (LN14 slides 19–22; LN15 slide 4)

The electrical and mechanical sides are coupled through the field:

- electrical: $v = Ri + \dfrac{d\lambda}{dt} = Ri + L(x)\dfrac{di}{dt} + i\dfrac{dL}{dx}\dfrac{dx}{dt}$ (the last term is the **speed voltage**);
- mechanical: $f_{fld} = M\dfrac{d^2x}{dt^2} + B\dfrac{dx}{dt} + k(x - x_0) + f_{ext}$ (LN14 slide 22), or rotationally $T_e = J\dfrac{d\omega}{dt} + B\omega + T_L$.

The spring stores energy without loss (analogue of an inductor, LN14 slide 20); the damper dissipates (analogue of a resistor, slide 21); the mass/inertia stores kinetic energy. The coupled equations are nonlinear in general and are solved numerically (Example 24.6). LN14 closes with applications: Bose's linear-motor active suspension and satellite attitude control by reaction wheels and magnetorquers (context only).

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6. Methods and common mistakes

1. Write $L_{11}(\theta)$, $L_{22}(\theta)$, $M(\theta)$ (given, or from reluctance arguments: periodic with π for saliency, 2π for mutual).

2. Coenergy (24.3) → torque (24.4); differentiate *before* substituting currents.
3. For AC excitation, substitute $i(t)$ and θ(t), use product-to-sum identities, keep only zero-frequency terms for the average.
4. Check the sign with physics: torque turns the rotor toward larger inductance / aligned axes.

Mistakes: forgetting the factor ½ on self-inductance terms but not on the mutual term; using θ-periodicity 2π for a salient rotor (it is π); claiming average torque at standstill for an AC-fed synchronous machine; treating rpm as rad/s.

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7. Worked examples

Example 24.1 — Reluctance rotor with DC (LN13 slides 7–11)

Given. $L(\theta) = 0.1 + 0.05\cos 2\theta$ H, $I = 5$ A DC. **Find** $T(\theta)$, its maximum, and T at 30°; check with the reluctance form. **Solution.** $T = -I^2 L_2 \sin 2\theta = -1.25 \sin 2\theta$ N·m; maximum 1.25 N·m at θ = ±45°; at 30°: −1.08 N·m (toward θ = 0). With $N = 100$ turns, $\mathcal{R}(\theta) = N^2/L(\theta)$ and $\Phi = NI/\mathcal{R}$; $-\frac{1}{2}\Phi^2\, d\mathcal{R}/d\theta$ evaluated numerically gives the same −1.08 N·m.

Example 24.2 — Reluctance motor on AC: when is there average torque? (Sen Example 3.5)

Given. Same rotor, $i = 10\sin(2\pi 50 t)$ A, δ = 0.3 rad. **Find** the average torque at standstill, at synchronous speed, and at 60 % of synchronous speed. **Solution.** From Section 3.2: standstill $-\frac{1}{2} I_m^2 L_2 \sin 2\delta = -1.41$ N·m (a holding torque toward alignment); synchronous ($\omega_m = 100\pi$ rad/s, 3000 rpm for this 2-pole rotor) $+\frac{1}{4} I_m^2 L_2 \sin 2\delta = +0.706$ N·m; at 0.6× synchronous: zero. The script confirms all three by numerical time-averaging.

Example 24.3 — Doubly excited salient machine (Final Fall 2014 Q2)

Given. (exams/archive/Final/Final - Fall2014.pdf p. 2.) $L_F = 0.01$ H (constant), $L_a(\theta) = 0.03 + 0.02\cos 2\theta$ H, $M_{aF}(\theta) = 0.25\cos\theta$ H. (a) Show $T = \frac{1}{2} i_F^2\, dL_F/d\theta + \frac{1}{2} i_a^2\, dL_a/d\theta + i_a i_F\, dM_{aF}/d\theta$. (b) With $\theta = 100\pi t + \delta$, $i_a = 10\cos 100\pi t$ A, $i_F = 0.5$ A, find T_{inst}. (c) Average torque at 100π rad/s. **Solution.** (a) From (24.3)–(24.4) with coenergy $W' = \frac{1}{2} L_F i_F^2 + \frac{1}{2} L_a i_a^2 + M_{aF} i_a i_F$.

b. $dL_F/d\theta = 0$; $dL_a/d\theta = -0.04\sin 2\theta$; $dM/d\theta = -0.25\sin\theta$:

$$T_{inst} = -2\cos^2(100\pi t)\sin(200\pi t + 2\delta) - 1.25\cos(100\pi t)\sin(100\pi t + \delta)\ \text{N}\cdot\text{m}.$$

c. Using $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$ and $\sin A\cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$, the averages are $\langle \cos^2(\omega t)\sin(2\omega t + 2\delta)\rangle = \frac{1}{4}\sin 2\delta$ and $\langle \cos\omega t \sin(\omega t + \delta)\rangle = \frac{1}{2}\sin\delta$:

$$T_{avg} = -0.625\sin\delta - 0.5\sin 2\delta\ \text{N}\cdot\text{m}.$$

(Checked by numerical averaging for δ = 0.2 and 0.5 rad: −0.319 and −0.720 N·m.) The first term is the mutual (synchronous) torque, the second the reluctance torque of the salient stator coupling.

Example 24.4 — Elementary cylindrical machine (Final Fall 2018 Q1)

Given. (exams/archive/Final/Final - Fall2018.pdf p. 2.) Smooth air gap: $L_s = L_1$, $L_r = L_2$ (constant), $L_{sr} = L_3\cos\theta_m$; rotor DC I_r, stator AC $i_s = I_m\cos\omega_e t$; $\theta_m = \omega_m t + \delta$. **Find** the torque expression, the speeds of non-zero average torque, and the average torque; numbers: $I_m = 10$ A, $I_r = 5$ A, $L_3 = 0.1$ H, 50 Hz. **Solution.** Only M varies: $T = i_s i_r\, dL_{sr}/d\theta = -I_m I_r L_3 \cos\omega_e t \sin(\omega_m t + \delta)$.

Product-to-sum: frequencies $\omega_m \pm \omega_e$ ⇒ average torque only at $|\omega_m| = \omega_e$ (synchronous speed, either direction). At $\omega_m = +\omega_e$: $T_{avg} = -\frac{1}{2} I_m I_r L_3 \sin\delta$ — maximum magnitude 2.5 N·m at δ = ∓90°; δ sets the sign (motoring or generating). At standstill the average is zero. With both windings on DC the rotor simply aligns ($T = -I_s I_r L_3 \sin\theta$); continuous rotation on DC requires switching one current with position — a commutator, i.e. the DC machine (Lesson 25; the exam's last part makes exactly this point).

Example 24.5 — Dyson-type rotors (Final Fall 2015 Q2)

Given. (exams/archive/Final/Final - Fall2015.pdf pp. 4–7.) Rotor #1: $L_{11} = L_a$ (constant), $L_{22} = L_b + L_c \cos 2\theta$, $M = M_{max} \cos\theta$. **Find** (a) the torque; (b) the motion for (i_1, i_2) = (1, 0), (0, 1), (1, 1) A DC. **Solution.** (a) $T = -L_c i_2^2 \sin 2\theta - M_{max} i_1 i_2 \sin\theta$ (SymPy-verified).

b. (1, 0): $T = 0$ — the stator alone sees a constant inductance, the rotor does not move. (0, 1): reluctance torque only, $-L_c \sin 2\theta$: the rotor turns to the nearest θ = 0 or π (two stable positions) and stops. (1, 1): both terms; the stable equilibrium is θ = 0 only (both terms restore there; at θ = π the mutual term is destabilizing if $M_{max} > 2L_c$). No continuous rotation with DC; Part III's switched-reluctance rotor needs current only while $dL/d\theta > 0$ in the direction of motion (switch on just after the unaligned position, off before alignment) — the principle of Dyson's digital motor.

Example 24.6 — Rotor dynamics under reluctance torque (original)

Given. The rotor of Example 24.1 ($I = 5$ A DC, $L_2 = 0.05$ H) with $J = 0.01$ kg·m² and viscous damping $B = 0.02$ N·m·s, released at rest from θ = 40°. **Find** the final position and compare the peak kinetic energy with the coenergy available. **Solution.** $J\ddot{\theta} = -I^2 L_2 \sin 2\theta - B\dot{\theta}$. Numerical integration: the rotor overshoots, oscillates and settles at θ = 0 (aligned, maximum inductance). Coenergy gained from 40° to 0°: $\frac{1}{2} I^2 (L(0) - L(40^\circ)) = 0.516$ J; the peak kinetic energy is 0.421 J — less, because the damper dissipates energy on the way (energy conservation check).

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8. Optional practice

P24.1. A rotor has $L(\theta) = 0.2 + 0.1\cos 2\theta$ H. What DC current gives a maximum holding torque of 5 N·m?

▼ **Solution**
$T_{max} = I^2 L_2 = 0.1 I^2 = 5$ ⇒ $I = 7.07$ A.

P24.2. For Example 24.3, find δ that maximizes the motoring torque magnitude if the machine should *motor* (positive average torque requires negative δ here).

▼ **Solution**
Maximize $g(\delta) = 0.625 \sin\delta' + 0.5 \sin 2\delta'$ with δ′ = −δ: $g' = 0.625\cos\delta' + \cos 2\delta' = 0$ ⇒ $2\cos^2\delta' + 0.625\cos\delta' - 1 = 0$ ⇒ $\cos\delta' = 0.568$ ⇒ δ′ = 55.4°, $T_{avg,max} = 0.625 \sin 55.4^\circ + 0.5 \sin 110.8^\circ = 0.982$ N·m at δ = −55.4°.

P24.3 (concept). Why can't a single-phase synchronous machine start by itself, and why do large machines use three phases?

▼ **Solution**
At standstill the torque oscillates at the supply frequency with zero average (Section 4.4), so the rotor only vibrates. Even at synchronous speed, a single-phase machine's torque pulsates

at twice the supply frequency, causing vibration and noise. A balanced three-phase winding produces a rotating MMF of constant amplitude and a constant torque (EE362).

9. Summary

Relation	Validity
$T = -\partial W_f/\partial\theta\|_\lambda = \partial W'_f/\partial\theta\|_i$; linear $T = \frac{1}{2}i^2\,dL/d\theta$	singly excited
$L = L_0 + L_2\cos 2\theta \Rightarrow T = -I^2 L_2 \sin 2\theta$ (DC)	salient rotor
AC reluctance motor: $T_{avg} = \frac{1}{4}I_m^2 L_2 \sin 2\delta$ at $\omega_m = \pm\omega$	synchronous speed
$W' = \frac{1}{2}\mathbf{I}^T\mathbf{L}\mathbf{I}$; $T = \frac{1}{2}\mathbf{I}^T(d\mathbf{L}/d\theta)\mathbf{I}$	multiply excited, linear
average torque only if $\|\omega_m\| = \|\omega_s \pm \omega_r\|$	AC currents
$T_e = J\dot{\omega} + B\omega + T_L$	mechanical side

10. Vocabulary

English	Türkçe
torque	moment (tork)
moment of inertia	eylemsizlik momenti
angular velocity	açısal hız
reluctance motor; switched-reluctance motor	relüktans motoru; anahtarlamalı relüktans motoru
salient pole; cylindrical (round) rotor	çıkık kutup; silindirik rotor
multiply excited system	çoklu uyarılmış sistem
reluctance torque; alignment (mutual) torque	relüktans momenti; hizalanma (ortak) momenti
synchronous speed; torque angle	senkron hız; moment (yük) açısı
damping; spring	sönümleme; yay

11. Sources

- LN13 viewer pp. 7–40 (slides 5–20). LN14 viewer pp. 3–52 (slides 2–32). Notes: LN13, LN14.
- Sen §3.4–3.5 pp. 108–113 (viewer 132–137): Eqs. 3.39–3.55, Example 3.5, Figs. 3.6–3.7.
- Final Fall 2014 Q2 p. 2; Final Fall 2015 Q2 pp. 4–7; Final Fall 2018 Q1 p. 2 (with typed hints/answers in the paper).
- Verification: `lesson24_torque_multiply_excited.py` (SymPy derivations, numerical averaging, ODE dynamics).
- More solved archive problems: Ch IV archive collection §7 (Final Fall 2018 Q4 concept items 4.5, 4.7).

Lesson 25 — DC machines: construction, commutation, generated EMF and electromagnetic torque

Chapter IV: Electromechanical Energy Converters Basics — DC Machines (DA Makineleri) · Week 11 · Lesson 25 of 31

Assistant-written teaching note. Teacher source: LN15 slides 1–31 (review, generators = motors, design of a DC motor, Lorentz force, parts, commutator, working principle, induced voltage in a rotating coil, rectified waveform, average EMF, ways to increase torque, torque smoothing, $E_a = K_a \omega_m \Phi_{pp}$). Book: Sen §4.1–4.2.6 (printed pp. 119–139, viewer 143–163), Example 4.1. Exam: Jan 2022 Final Q3 Part I (parts of a DC machine). Numbers: `lesson25_26_dc_machines.py`.

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1. Place and goals

The DC machine is the first complete rotating electromechanical converter of the course. “Generators = motors: there is no difference between a DC motor and generator” (LN15 slides 5, 30) — the same machine converts in either direction depending on the operating point. It brings together everything in Chapters II and IV: a magnetic circuit with an air gap sets the flux, Lorentz force on armature conductors gives torque, the motional EMF gives the back EMF, and a mechanical switch — the commutator — turns the alternating coil quantities into DC at the terminals.

You should be able to:

1. Name the parts of a DC machine and their functions, and say which rotate.
2. Derive the EMF of a coil rotating in a magnetic field and explain how the commutator rectifies it.
3. Derive $E_a = K_a \Phi \omega_m$ and $T = K_a \Phi I_a$ from conductor-level physics, and show $E_a I_a = T \omega_m$.
4. Compute the machine constant for lap and wave windings.
5. Explain the magnetization curve and the ways to increase torque, and why many coils smooth the torque.

Prerequisites: Lessons 12–13 (magnetic circuits with gaps), 22 (Lorentz force, motional EMF), 24 (alignment torque and why DC needs commutation).

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2. Construction (LN15 slides 11–19; Sen §4.2.1)

Part	Turkish	Where	Function
stator / frame (yoke)	stator / gövde (boyunduruk)	stationary	mechanical support, return path for the field flux
field poles and pole shoes	alan kutupları, kutup pabuçları	stationary	carry the field winding; shape the gap flux
field winding (or permanent magnets)	alan sargısı	stationary	DC excitation producing the main flux Φ per pole

Part	Turkish	Where	Function
rotor / armature core	rotor / endüvi nüvesi	rotates	laminated slotted iron (its flux alternates as it turns)
armature winding	endüvi sargısı	rotates	conductors in slots in which EMF is induced and torque produced
commutator	komütatör (kolektör)	rotates	copper segments insulated from each other, connected to the armature coils
brushes	fırçalar	stationary	carbon blocks sliding on the commutator, connecting the rotating winding to the terminals
shaft, bearings	mil, rulmanlar	rotate/support	mechanical port

(Jan 2022 Final Q3 Part I asks exactly this naming and "rotates: yes/no".) The armature core is laminated because, seen from the rotor, the field reverses every time a conductor passes from a N to a S pole (Lesson 16). The field winding of a *shunt* or *separately excited* machine has many turns of thin wire (small current); a *series* field has few turns of thick wire (Sen §4.2.7). A permanent-magnet DC (PMDC) machine replaces the field winding by magnets: a separately excited machine with fixed flux (LN15 slide 28).

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3. Working principle (LN15 slides 16–20)

Motor action: current fed through the brushes and commutator flows in the armature conductors; each conductor in the pole field feels $F = BIl$ (Lesson 22), and the forces on the two sides of a coil form a torque about the shaft. **Generator action:** a prime mover turns the armature; each conductor moving through the field has an EMF $e = Blv$.

Both happen simultaneously in any loaded machine (Sen §4.1): a motor's conductors also generate a **back EMF** opposing the supply, and a generator's current produces a **counter-torque** opposing the prime mover. Which effect "wins" — whether electrical power flows in or out — is set by the operating point (Lesson 26).

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4. EMF of a single rotating coil and commutation

4.1 Flux through the coil (LN15 slides 17–19)

Take a rectangular coil of N turns, length L (along the shaft) and radius r (half-width r, so area $2rL$), rotating at ω in a uniform field $\hat{B}$ (the geometry of Figure 22.1). The flux linked is maximum when the coil plane is perpendicular to **B**:

$$\Phi_{max} = \hat{B}(2r)L = 2\hat{B}rL \qquad \text{(LN15 slide 19)}.$$

As the coil turns, $\Phi(t) = \Phi_{max}\cos\omega t$, and with Faraday's law (LN15's sign convention for the EMF):

$$v_{coil} = -N\frac{d\Phi}{dt} = N\omega\Phi_{max}\sin\omega t. \tag{25.1}$$

The coil voltage is **alternating** — a rotating coil is naturally an AC generator.

4.2 The commutator as a mechanical rectifier (LN15 slides 20–21)

The two ends of the coil are connected to two commutator segments; the brushes touch the segment that is currently "positive". Every half turn, exactly when the coil EMF passes through zero, the segments swap brushes. The brush voltage is therefore $|v_{coil}|$ (Figure 25.1, left): "mechanical brushes work as a diode rectifier". Its average, for a two-pole machine, is

$$E_{av} = \frac{2}{\pi}E_{max} = \frac{2}{\pi}N\omega\Phi_{max} \qquad \text{(LN15 slide 21)}. \tag{25.2}$$

In motor operation the commutator works the other way round (as an inverter): it reverses the coil current every half turn so that the torque on the coil always has the same direction — the fix for the alignment problem of Lesson 24 (Example 24.4).

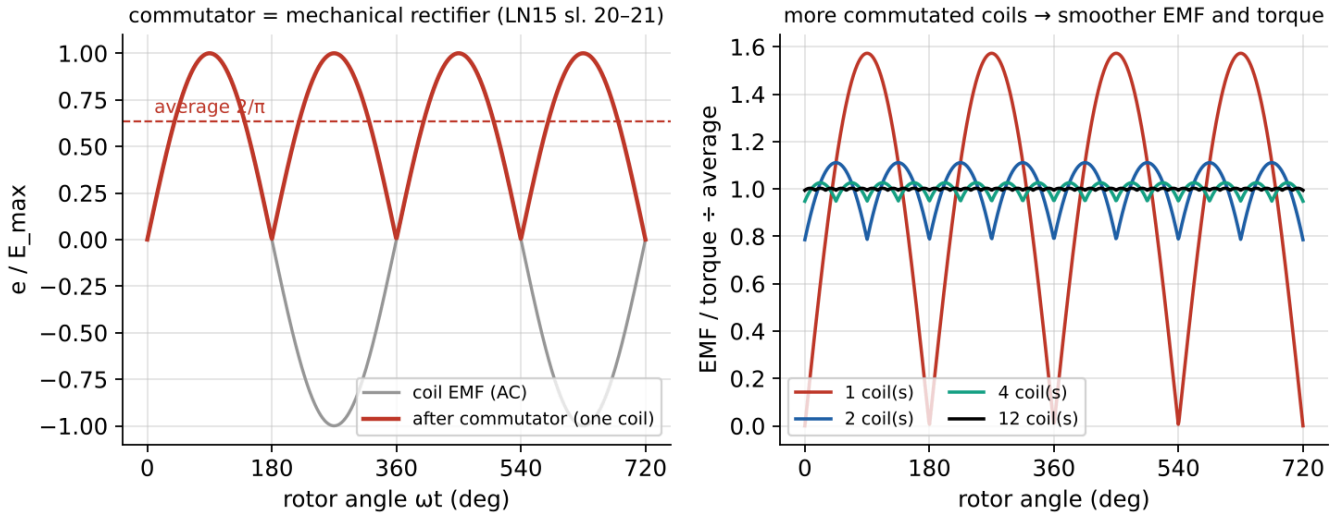


Figure 25.1. Left: coil EMF and the rectified brush voltage of one coil. Right: with k coils displaced around the armature, each commutated, the sum is far smoother (normalized; peak-to-peak ripple 157 % for one coil, 33 % for two, 7.8 % for four, 0.9 % for twelve).

4.3 Many coils: smooth EMF and torque (LN15 slides 25–26)

A single coil gives a pulsating EMF and, in a motor, a pulsating torque $\propto |\sin\omega t|$ (LN15: "can you plot the torque waveform?"). Real armatures have many coils in many slots, connected in series through many commutator segments, each coil displaced in angle. At every instant most coils are under a pole producing near-maximum EMF and torque, so the sum is nearly constant ("how can you smooth the torque output?" → distribute many coils around the armature).

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5. The EMF and torque equations

5.1 Generated EMF (Sen §4.2.4)

Let the machine have p poles, flux Φ per pole, Z armature conductors ($N = Z/2$ turns) arranged in a parallel paths between the brushes, and mechanical speed ω_m. The average flux density under the poles is $B = \Phi/A_{pole}$ with $A_{pole} = 2\pi rl/p$. Each conductor moves at $v = \omega_m r$, so its

average EMF is

$$\bar{e}_c = Blv = \frac{\Phi p}{2\pi r l}\, l\, \omega_m r = \frac{p\Phi\omega_m}{2\pi}.$$

Each parallel path has Z/a conductors in series:

$$\boxed{E_a = \frac{Z}{a}\cdot\frac{p\Phi\omega_m}{2\pi} = K_a\Phi\omega_m, \qquad K_a = \frac{Zp}{2\pi a} = \frac{Np}{\pi a}} \tag{25.3}$$

(Sen Eqs. 4.9–4.11; LN15 slides 29, 31: $E_a = K_a\omega_m\Phi_{pp}$, “induced voltage proportional to speed”). E_a is called the *generated voltage* in a generator and the *back EMF* (zıt EMK) in a motor — same formula.

5.2 Electromagnetic torque (Sen §4.2.5; LN15 slides 37–38)

Each conductor carries I_a/a; its average force is BlI_a/a and its torque $\bar{T}_c = Bl(I_a/a)r = \Phi p I_a/(2\pi a)$. All Z conductors add:

$$\boxed{T = K_a\Phi I_a} \qquad (\text{LN15 slide 37: } T = K_a\Phi_{pp}I_a;\ \text{”torque proportional to armature current”}). \tag{25.4}$$

5.3 Power balance

Multiplying: $E_aI_a = K_a\Phi\omega_m I_a = T\omega_m$ (Sen Eq. 4.17a; LN15 slide 37). The **electromechanical power** E_aI_a is the power converted between electrical and mechanical form (LN15 slides 35–36). The same constant $K_a\Phi$ appears in both equations: in SI units, $K_a\Phi$ in V·s/rad equals the torque constant in N·m/A — a useful consistency check on test data (Lesson 26, Example 26.6).

Lap and wave windings. In a lap winding $a = p$; in a wave winding $a = 2$ (Sen §4.2.3). For the same conductors, a wave winding gives fewer, longer paths: higher voltage, lower current, same power (Example 25.2).

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6. The magnetization curve (Sen §4.2.6)

The flux per pole is set by the field MMF and the reluctance of the magnetic circuit: pole, air gap, rotor teeth, rotor core, return through the yoke (Lesson 12 applied to a machine). While the iron is unsaturated, the gap dominates and $\Phi \approx \mathcal{F}_p/\mathcal{R}_g$ rises linearly with field current; at higher excitation the teeth saturate and Φ levels off; at zero field current a small **residual flux** remains (Lesson 11). Since $E_a \propto \Phi$ at constant speed, the curve is usually plotted as open-circuit E_a versus I_f at a stated speed (Sen Figs. 4.23–4.24) — the **magnetization (open-circuit) characteristic** used in Lesson 26. At another speed, scale the voltage axis by the speed ratio.

Ways to increase torque (LN15 slides 22–24): more flux density (stronger magnets or less reluctance — but iron saturates), more current (limited by heating/cooling), longer conductors (more force per conductor), larger radius (more torque per force). Torque per volume is roughly fixed by the magnetic and electric loadings, which is why high-torque machines are large.

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7. Methods and common mistakes

- Convert rpm to rad/s before using $E_a = K_a\Phi\omega_m$; or work with $K_a\Phi$ in V/rpm consistently (Sen Ex. 4.11).
- Count conductors Z (two per turn), not turns, in $K_a = Zp/(2\pi a)$; or use N turns with $K_a = Np/(\pi a)$.

- The armature current splits into a paths; conductor current is I_a/a.
- E_a is internal; terminal voltage differs by I_aR_a (Lesson 26).

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8. Worked examples

Example 25.1 — A single coil generator (original, LN15 slides 17–21)

Given. A 20-turn coil, $r = 5$ cm, $L = 12$ cm, in $\hat{B} = 0.6$ T, at 1500 rpm. **Find** Φ_{max}, E_{max} and the average brush voltage with a commutator. **Solution.** $\Phi_{max} = 2(0.6)(0.05)(0.12) = 7.2$ mWb; ω = 157.1 rad/s; $E_{max} = N\omega\Phi_{max} = 22.6$ V; $E_{av} = (2/\pi)(22.6) = 14.4$ V. (Numerical averaging of $|\sin|$ confirms.)

Example 25.2 — Lap- versus wave-wound armature (Sen Example 4.1)

Given. Four-pole machine, armature radius 12.5 cm, effective length 25 cm, poles cover 75 % of the periphery, 33 coils × 7 turns in 33 slots, average pole flux density 0.75 T, 1000 rpm. Coil current rating 100 A. **Find** for (1) lap and (2) wave winding: K_a, E_a, torque at rated coil current, power. **Solution.** $Z = 2 \times 33 \times 7 = 462$. Pole area $= 2\pi(0.125)(0.25)(0.75)/4 = 0.0368$ m² ⇒ Φ = 0.0276 Wb; $\omega_m = 104.7$ rad/s.

1. Lap, $a = p = 4$: $K_a = 462 \times 4/(2\pi \times 4) = 73.5$; $E_a = 73.5(0.0276)(104.7) = 212.6$ V (Sen 212.5 V with Φ rounded); $I_a = 4 \times 100 = 400$ A; $T = K_a\Phi I_a = 812$ N·m; $P = E_aI_a = 85.0$ kW $= T\omega_m$ ✓.
2. Wave, $a = 2$: $K_a = 147.1$; $E_a = 425$ V; $I_a = 200$ A; $T = 812$ N·m; $P = 85.0$ kW. **Interpretation.** The winding type trades voltage against current; torque and power for the same conductor current are identical.

Example 25.3 — Torque ripple and the number of coils (LN15 slides 25–26)

For k identical commutated coils spaced π/k apart, the summed rectified waveform has peak-to-peak ripple (relative to the average) of 157 % (k = 1), 33 % (k = 2), 7.8 % (k = 4), 0.86 % (k = 12) (computed in the script; Figure 25.1). Practical armatures with dozens of segments give essentially constant torque.

Example 25.4 — Naming the parts (Jan 2022 Final Q3 Part I)

From the typical cut-away (LN15 slide 14): armature (rotor, rotates: yes), commutator (rotates: yes), brushes (stationary), field winding and pole pieces (stationary), frame/yoke (stationary). The brushes are stationary because they connect the external (stationary) circuit to the rotating commutator.

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9. Optional practice

P25.1. A 6-pole, lap-wound armature has 720 conductors; flux per pole 25 mWb. Find E_a at 1000 rpm and the torque at $I_a = 100$ A.

▼ **Solution**
$a = 6$, $K_a = 720 \times 6/(2\pi \times 6) = 114.6$; $E_a = 114.6 \times 0.025 \times 104.7 = 300$ V; $T = 114.6 \times 0.025 \times 100 = 286.5$ N·m; check $E_aI_a = 30.0$ kW $= T\omega_m$.

P25.2. The same machine is rewound wave. What changes at the same speed and flux?

▼ **Solution**

$a = 2$: K_a triples (343.8) ⇒ $E_a = 900$ V; for the same conductor current (100/6 A) the armature current is 33.3 A; torque unchanged (286.5 N·m).

P25.3 (concept). A PMDC motor's no-load speed rises when the magnets weaken (e.g. overheating). Why?

▼ **Solution**
At no load $E_a \approx V_t \Rightarrow \omega_m \approx V_t/(K_a\Phi)$: smaller Φ requires a higher speed to generate the same back EMF. (Torque per ampere falls, so the motor draws more current for the same load.)

10. Summary

Relation	Validity
$\Phi_{max} = 2\hat{B}rL$; $v = N\omega\Phi_{max}\sin\omega t$	single coil, uniform B
$E_{av} = (2/\pi)E_{max}$	one commutated coil, two poles
$E_a = K_a\Phi\omega_m$, $K_a = Zp/(2\pi a)$	any DC machine
$T = K_a\Phi I_a$; $E_aI_a = T\omega_m$	electromagnetic (before losses)
lap $a = p$; wave $a = 2$	windings

11. Vocabulary

English	Türkçe
DC machine; DC generator / motor	doğru akım makinesi; DA generatör / motor
armature; armature winding	endüvi; endüvi sargısı
field winding; field pole; pole shoe	alan (uyartım) sargısı; alan kutbu; kutup pabucu
commutator; brush	komütatör (kolektör); fırça
yoke / frame	boyunduruk / gövde
generated EMF; back EMF	üretilen EMK; zıt EMK
electromagnetic torque	elektromanyetik moment
machine (armature) constant	makine (endüvi) sabiti
lap winding / wave winding	paralel (sarmal) sargı / seri (dalga) sargı
magnetization curve; residual flux	mıknatıslanma eğrisi; artık akı
permanent-magnet DC motor	sürekli mıknatıslı DA motoru

12. Sources

- LN15 viewer pp. 3–53 (slides 2–31). Notes.
- Sen §4.1–4.2.6 pp. 119–139 (viewer 143–163): Eqs. 4.1–4.18, Example 4.1, Figs. 4.19–4.24.
- Jan 2022 Final Q3 Part I (exams/archive/Final/EE361_2022_Final_Exam_Q4_solution.pdf p. 9).
- Verification: `lesson25_26_dc_machines.py`.
- More solved archive problems: Ch IV archive collection §1, §7 (Final Fall 2017 Q2 Part II coil/commutator waveforms; Final Fall 2018 Q4 concepts).

Lesson 26 — DC machine equivalent circuits, power flow, and separately excited generators and motors

Chapter IV: Electromechanical Energy Converters Basics — DC Machines (DA Makineleri) · Weeks 11–12 · Lesson 26 of 31

Assistant-written teaching note. Teacher source: LN15 slides 27–44 (simplest equivalent circuit, E_a, motor/generator action, separately excited equivalent circuit, motoring vs generating condition, copper losses, electromechanical power, torque relations, start-up, summary). Book (supplement beyond LN15, labelled): Sen §4.2.7, §4.3.1–4.3.2, §4.4.1, §4.4.3, §4.5 (printed pp. 139–183, viewer 163–207), Examples 4.2, 4.6, 4.10, 4.11. Exams: Final Fall 2014 Q3–Q4; Final Fall 2015 Q3–Q4; Final Fall 2018 Q2. Numbers: `lesson25_26_dc_machines.py`.

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1. Place and goals

Lesson 25 gave the two machine equations $E_a = K_a \Phi \omega_m$ and $T = K_a \Phi I_a$. Add the armature resistance and a field circuit and you have a complete steady-state model of a DC machine. The syllabus (Week 12) asks for **separately excited DC generators and motors**: how terminal voltage varies with load in a generator, how speed varies with torque in a motor, what happens at start-up, how speed is controlled, and how power flows and is lost. LN15 covers the model and start-up; the characteristics, starter design and speed control follow Sen Chapter 4 and the archived finals, and are labelled as such. Self-excited (shunt) generators appear in Final 2014 Q3 and Final 2015 Q3 and are treated briefly as an extension.

You should be able to:

1. Draw and write the equivalent circuit of a separately excited machine with consistent motor/generator sign conventions.
2. Use the magnetization curve with speed scaling, and compute generator terminal voltage under load.
3. Derive the motor speed–torque characteristic and compute speed, current, torque, power and efficiency at any operating point.
4. Explain and calculate the start-up current and design a resistance starter.
5. Use armature-voltage and field control, identify motoring/regenerating quadrants, and track energy during speed changes.

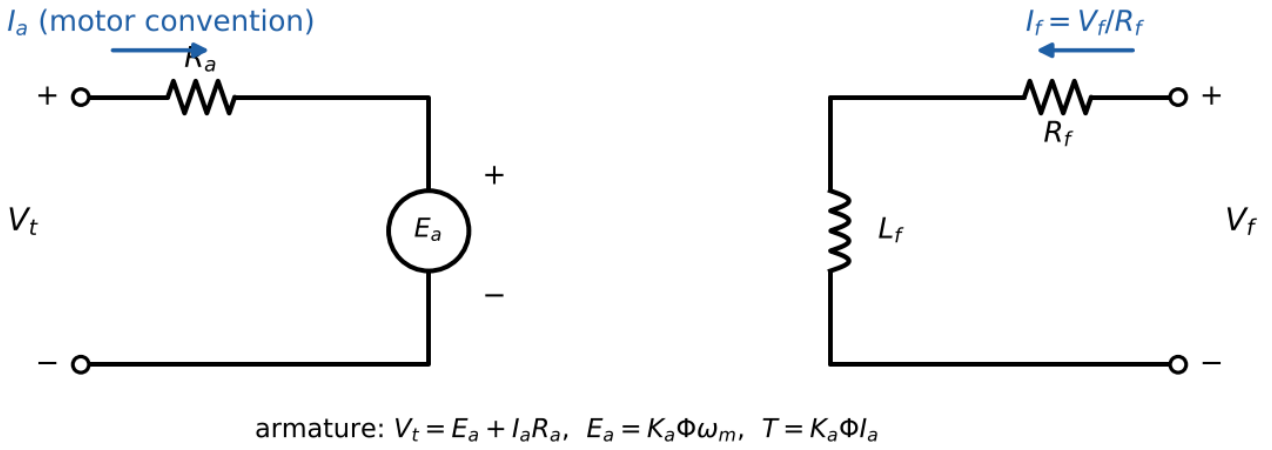
Prerequisites: Lesson 25; Chapter I DC power (Lesson 1).

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2. Equivalent circuit and conventions

2.1 The model (LN15 slides 28, 32–33; Sen Fig. 4.28)

Armature circuit: the generated EMF E_a in series with the armature resistance R_a (winding plus brushes; sometimes a separate brush drop of about 2 V is quoted, Sen p. 142). **Field circuit:** a separate DC source V_f drives $I_f = V_f / R_f$ through the field resistance R_f (winding plus rheostat) and inductance L_f (which matters only in transients). In steady state the inductances do not appear (Figure 26.1).



Motor convention (I_a flowing *into* the + terminal):

$$V_t = E_a + I_a R_a, \qquad E_a = K_a \Phi \omega_m, \qquad T = K_a \Phi I_a. \tag{26.1}$$

LN15 slide 33: if $V_t > E_a$ the current flows in and the machine motors; if $E_a > V_t$ the current reverses and the machine generates. In generator problems it is natural to define the current *out of* the + terminal (I_a = load current): $V_t = E_a - I_a R_a$ (Sen Eq. 4.24). Both are the same equation with the sign of I_a changed — choose one and state it.

2.2 Power flow and losses (LN15 slides 34–38; Sen Fig. 4.49)

Motor: electrical input $V_t I_a$ (+ field input $V_f I_f$) → minus armature copper loss $I_a^2 R_a$ → **electromechanical power** $P_e = E_a I_a = T\omega_m$ → minus rotational losses (friction, windage, rotor core loss) → shaft output.

$$P_e = V_t I_a - I_a^2 R_a = E_a I_a = T\omega_m. \tag{26.2}$$

(LN15 slide 35 prints $V_a I_a - I_a R_a$; the armature loss is $I_a^2 R_a$ — errata E7.) Copper losses: $P_{cu} = I_a^2 R_a + I_f^2 R_f$ (slide 34). Efficiency $\eta = P_{out}/P_{in}$, with the field power counted in the input for a self-contained machine. Generator: the arrows reverse — mechanical input → rotational loss → $E_a I_a$ → armature copper loss → electrical output $V_t I_a$.

Rotational loss from a no-load test (Sen Ex. 4.6): at no load the motor develops just enough torque to overcome rotation losses, so $P_{rot} \approx E_a I_{a,NL}$ at that speed.

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3. Separately excited generator (Sen §4.3.1)

No load: $V_t = E_a = K_a \Phi \omega_m$ — read from the magnetization curve at the given I_f, scaled by speed: $E_a(\omega_2) = E_a(\omega_1)\,\omega_2/\omega_1$ (same I_f).

Under load: $V_t = E_a - I_a R_a$, a slightly drooping straight line (the **external characteristic**, Figure 26.2 middle); intersect it with the load line $V_t = I_a R_L$ to find the operating point: $I_a = E_a/(R_a + R_L)$.

Armature reaction (Sen p. 143): at large current the armature MMF distorts the gap flux; with saturation the flux per pole falls, causing an additional voltage drop. It is modelled as an equivalent reduction of field current, $I_{f,eff} = I_f - I_{f(AR)}$. Neglect it unless data are given.

Ways to restore the terminal voltage under load: raise the field current (so E_a rises by $I_a R_a$) or raise the speed (Final 2015 Q3).

Self-excited (shunt) generator — extension (Sen §4.3.2). The field is fed from the armature itself. Voltage builds up from residual magnetism and settles where the magnetization curve meets the field-resistance line $V = R_f I_f$ (Figure 26.2 left). Conditions: residual flux present, field connected so it aids it, and R_f below the critical resistance (slope of the linear part of the curve). Under load the terminal voltage falls more than in the separately excited case because the field current falls with V_t.

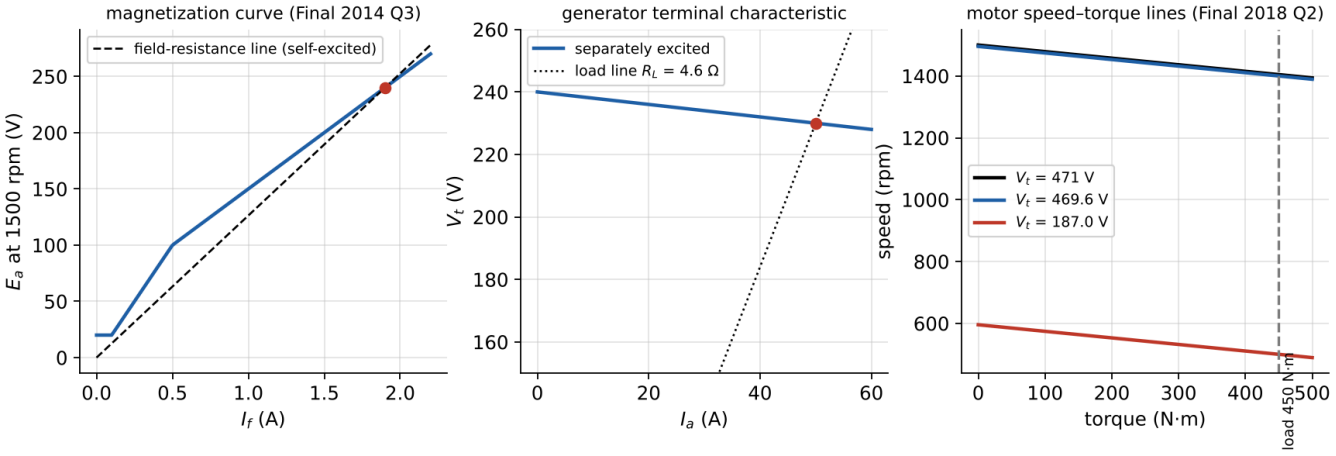


Figure 26.2. Left: magnetization curve of Final 2014 Q3 with the self-excitation field line. Middle: separately excited generator terminal characteristic and load line. Right: motor speed–torque lines for three armature voltages (Final 2018 Q2 machine).

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4. Separately excited motor (LN15 slides 32–43; Sen §4.4.1)

4.1 Speed–torque characteristic

From (26.1) with constant field:

$$\omega_m = \frac{E_a}{K_a\Phi} = \frac{V_t - I_a R_a}{K_a\Phi} = \frac{V_t}{K_a\Phi} - \frac{R_a}{(K_a\Phi)^2}\,T \qquad \text{(Final 2015 Q4 Part I).} \tag{26.3}$$

A straight line: the **no-load speed** $\omega_0 = V_t/(K_a\Phi)$ minus a small droop proportional to torque. Because R_a is small, speed drops only a few percent from no load to full load — the separately excited (and shunt) motor is nearly a constant-speed motor. The operating point is where the motor torque equals the load torque; the motor draws whatever current that torque requires: $I_a = T_L/(K_a\Phi)$.

4.2 Start-up (LN15 slides 39–42; Sen §4.4.3)

At standstill $\omega_m = 0 \Rightarrow E_a = 0 \Rightarrow I_{a,start} = V_t/R_a$ — often 10–20 times rated current, with a correspondingly huge starting torque (“never attempt a drag race with an electric car”, LN15 slide 40). The current must be limited:

1. **Starting resistance** R_{ae} in series with the armature, removed in steps as the speed (and back EMF) builds up (Example 26.4; Sen Fig. 4.57);
2. **Reduced armature voltage** from a controllable converter, raised as the motor accelerates (Chapter V, Sen Ex. 4.11).

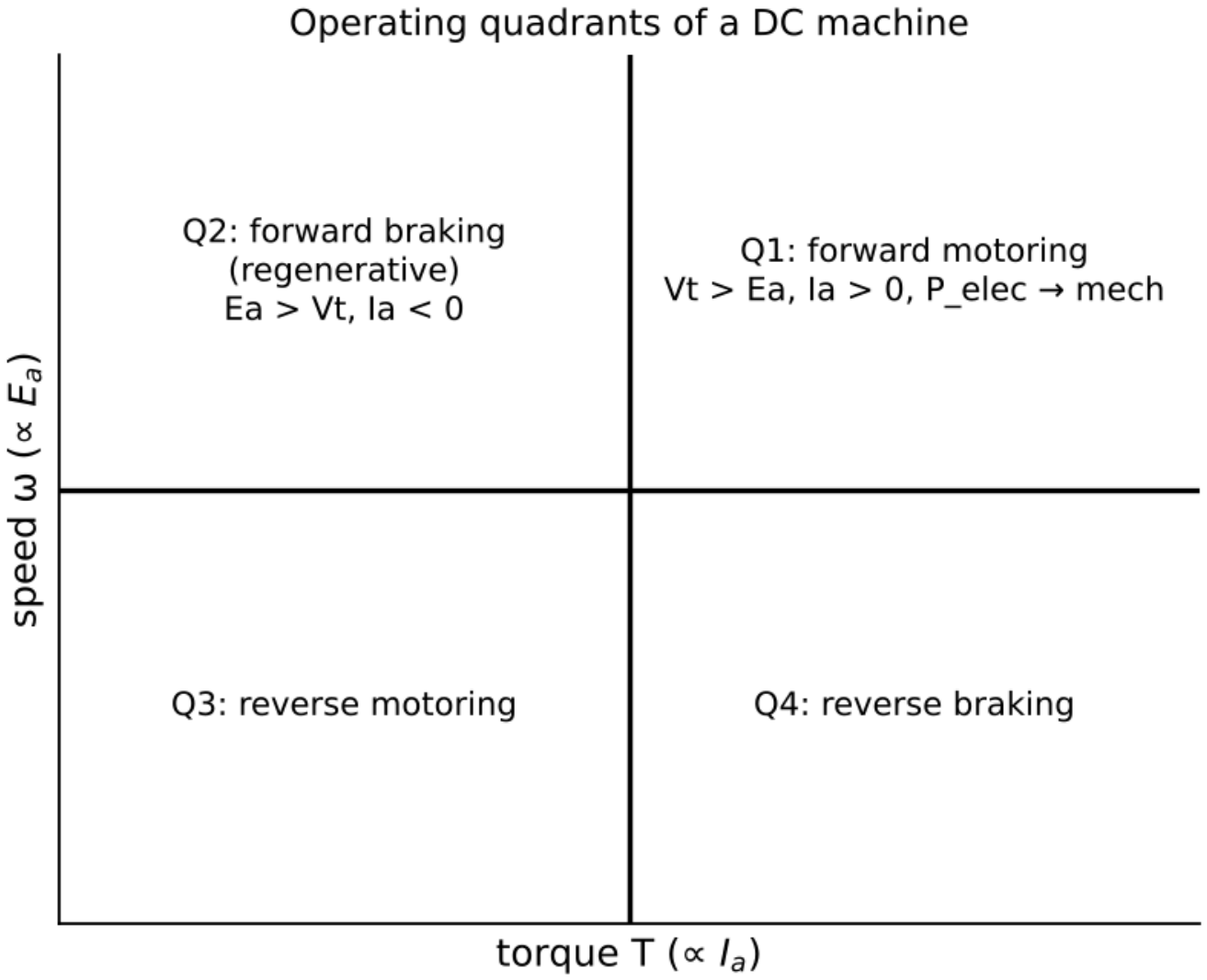
4.3 Speed control (Sen §4.5 — supplement)

From (26.3), speed can be changed by:

- **Armature voltage control** (constant field): speed ∝ V_t approximately; full torque available at every speed (constant-torque region below base speed). Classic Ward–Leonard motor–generator sets, today a controlled rectifier or chopper.
- **Field weakening** (constant V_t): reducing I_f reduces Φ and *raises* speed above base speed; available torque per ampere falls (constant-power region).
- **Armature resistance**: wasteful, used mainly for starting.

4.4 Four-quadrant operation and regenerative braking

With a converter that can reverse current (and voltage), the machine can motor or brake in either direction (Figure 26.3). If V_t is reduced below E_a while the machine is turning, I_a reverses: the machine **generates**, its torque opposes motion and decelerates the drive, and the kinetic energy is returned to the supply (regenerative braking) — Final 2015 Q4 Part III and Final 2018 Q2 Part III. Connecting a resistor instead dissipates that energy (dynamic braking, Final 2014 Q4(c)).



5. Methods and common mistakes

- Find $K_a\Phi$ from a known operating point: $K_a\Phi = E_a/\omega_m$ (no-load: $E_a \approx V_t$ if friction is negligible).
- If the field is constant, $K_a\Phi$ is constant: $E_a \propto \omega_m$ and $T \propto I_a$. If the field changes, use ratios: $E_{a2}/E_{a1} = (\Phi_2/\Phi_1)(\omega_2/\omega_1)$.
- In SI, $K_a\Phi$ (V·s/rad) = torque constant (N·m/A); in V/rpm multiply by $60/(2\pi)$ to get N·m/A.
- Keep the sign convention fixed; a negative I_a in motor convention means generating.
- Rotational losses are subtracted from E_aI_a to get shaft power (motor) or added to it to get prime-mover power (generator).

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6. Worked examples

Example 26.1 — Separately excited generator under load (Sen Example 4.2)

Given. 12 kW, 100 V, 1000 rpm machine, $R_a = 0.1\ \Omega$, rated field current 1 A (E_a = 100 V at 1000 rpm). Separately excited at 1000 rpm, rated field current. **Find** (a) full-load terminal voltage without armature reaction; (b) with armature reaction equivalent to 0.06 field-amperes: (i) full-load V_t; (ii) field current for $V_t = 100$ V at full load. **Solution.** Full-load current 12 000/100 = 120 A. (a) $V_t = 100 - 120(0.1) = 88$ V. (b)(i) $I_{f,eff} = 0.94$ A ⇒ $E_a = 98$ V (Sen's curve) ⇒ $V_t = 86$ V. (ii) Need $E_a = 100 + 12 = 112$ V ⇒ $I_{f,eff} = 1.4$ A (curve) ⇒ $I_f = 1.46$ A.

Example 26.2 — Generator with a piecewise magnetization curve (Final Fall 2014 Q3)

Given. (exams/archive/Final/Final - Fall2014.pdf p. 3.) Two-pole generator at 1500 rpm: residual 20 V; $E_a = 200I_f$ for $I_f \le 0.5$ A; $E_a = 100I_f + 50$ for $I_f > 0.5$ A. Field circuit total 126.116 Ω; $R_a = 0.2\ \Omega$. Separately excited with $I_f = 1.9$ A. **Find** (a) E_a and V_t at no load; (b) with $R_L = 4.6\ \Omega$: I_a, V_t, P_{out}. Extension (c)–(e): shunt-connected at no load and with the same load. **Solution.** (a) $E_a = 100(1.9) + 50 = 240$ V $= V_t$. (b) $I_a = 240/(0.2 + 4.6) = 50$ A; $V_t = 50(4.6) = 230$ V; $P_{out} = 11.5$ kW.

c. Self-excited, no load: field line $V = 126.116I_f$; with the armature drop of the field current included, $100I_f + 50 = (126.116 + 0.2)I_f$ ⇒ $I_f = 1.900$ A, $E_a = 240.0$ V, $V_t = 239.6$ V (the field resistance was chosen to reproduce 240 V). (d, e) With 4.6 Ω: $V_t = 126.116I_f$ and $E_a = V_t + (I_f + V_t/4.6)(0.2)$ ⇒ $I_f = 1.572$ A, $E_a = 207.2$ V, $V_t = 198.3$ V, $I_L = 43.1$ A, $P_{out} = 8.55$ kW. **Comparison:** the self-excited generator's voltage falls by 17 % versus 4 % for the separately excited one, because its field current falls with the terminal voltage.

Example 26.3 — Shunt-connected motor: no-load test, full-load performance (Sen Example 4.6)

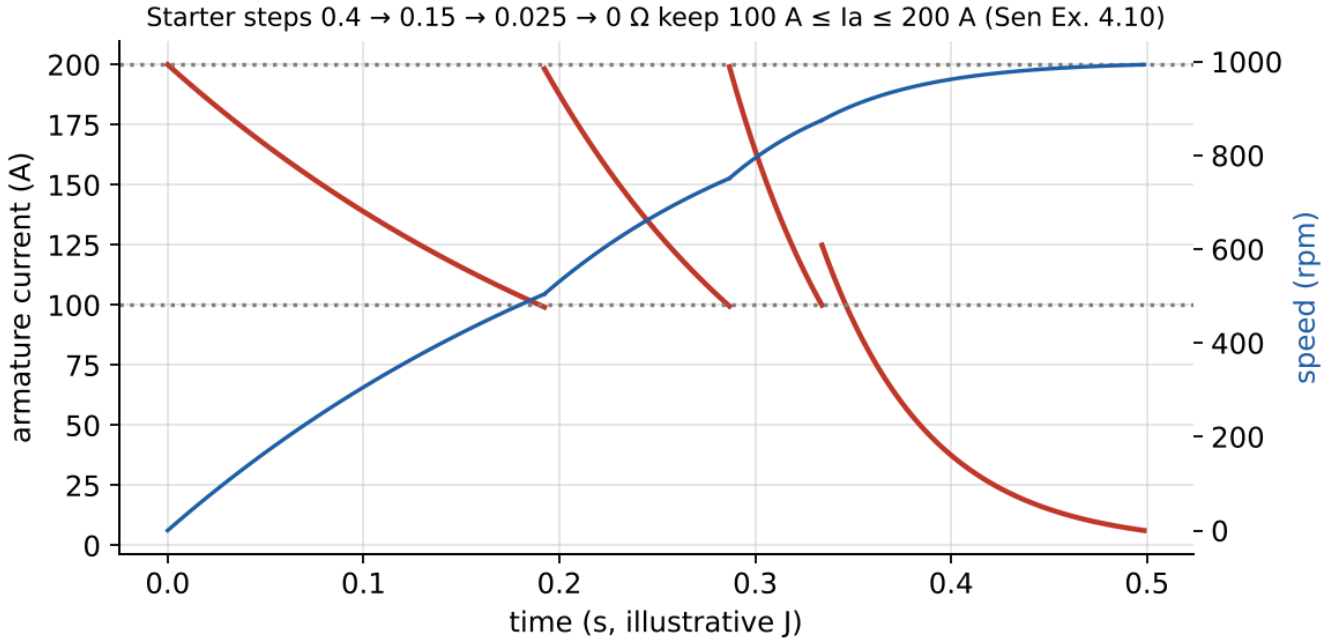
Given. Same 12 kW, 100 V machine as a motor on 100 V; at no load 1000 rpm and I_a = 6 A; field 0.99 A. **Find** rotational loss; speed, torque and efficiency at rated armature current (120 A) (i) without, (ii) with a 5 % flux reduction; starting torque at 150 % current. **Solution.** $E_{a,NL} = 100 - 0.6 = 99.4$ V; $P_{rot} = 99.4 \times 6 = 596$ W; $K_a\Phi = 99.4/104.72 = 0.949$ V·s/rad.

i. $E_{a,FL} = 100 - 12 = 88$ V ⇒ $n = 1000(88/99.4) = 885$ rpm; $T = E_aI_a/\omega_m = 10\,560/92.7 = 113.9$ N·m; $P_{out} = 10\,560 - 596 = 9964$ W; $P_{in} = 100(120.99) = 12\,099$ W; η = 82.35 %.
ii. 5 % less flux ⇒ $n = 885/0.95 = 932$ rpm (flux reduction *raises* speed); $T = 108.2$ N·m. Start at 180 A: $T = 0.949 \times 180 = 171$ N·m (without armature reaction).

Example 26.4 — Designing a resistance starter (Sen Example 4.10)

Given. 10 kW, 100 V, 1000 rpm, $R_a = 0.1\ \Omega$ on 100 V. **Find** (a) direct-on-line starting current; (b) starting resistance for 2 × rated current; (c) starter steps keeping I_a between 1 and 2 pu. **Solution.** Rated current 100 A. (a) 100/0.1 = 1000 A = 10 pu. (b) $100/(0.1 + R) = 200 \Rightarrow R = 0.4\ \Omega$.

c. Step 1: external 0.4 Ω (total 0.5 Ω); the current decays to 100 A when $E_a = 100 - 100(0.5) = 50$ V. Switch so the current jumps back to 200 A: $R_2 = (100 - 50)/200 - 0.1 = 0.15\ \Omega$. Next at 100 A: $E_a = 75$ V ⇒ $R_3 = 0.025\ \Omega$. Next: $E_a = 87.5$ V ⇒ required $R_4 < 0$ ⇒ cut out all resistance (current jumps only to 125 A). Sections: 0.25, 0.125, 0.025 Ω (Figure 26.4).



Example 26.5 — Constant-torque drive with armature-voltage control (Final Fall 2018 Q2)

Given. (exams/archive/Final/Final - Fall2018.pdf pp. 4–5.) Separately excited motor fed by a fully controllable, bidirectional converter; constant field; $R_a = 0.2\ \Omega$; no-load 1500 rpm at 471 V; friction neglected; constant-torque load 450 N·m; $J = 2$ kg·m². **Find** (a) E_a at no load; (b) V_t, I_a at 1400 rpm with the load; (c) gross mechanical power; (d) V_t at 500 rpm; (e) stored kinetic energy at 1400 rpm; (f) energy to extract when slowing to 500 rpm; (g) what happens to it. **Solution.** (a) No friction ⇒ $I_a = 0$ at no load ⇒ $E_a = 471$ V; $K_a\Phi = 471/157.08 = 2.998$ V·s/rad. (b) $I_a = 450/2.998 = 150.1$ A (rated); $E_a = 2.998(146.6) = 439.6$ V; $V_t = 439.6 + 150.1(0.2) = 469.6$ V. (c) $P_m = T\omega_m = 450 \times 146.6 = 65.97$ kW. (d) $E_a = 2.998(52.36) = 157.0$ V; $V_t = 187.0$ V. (e) $\frac{1}{2}J\omega^2 = 21.49$ kJ. (f) $\Delta KE = 21.49 - 2.74 = 18.75$ kJ.

g. To decelerate, the converter lowers V_t below E_a: I_a reverses, the machine generates and its torque adds to the load torque. *Quantified extension:* braking at rated current (−150 A) gives a deceleration time of 0.21 s; of the 18.75 kJ released, 9.38 kJ still goes into the load (which keeps demanding 450 N·m), 9.37 kJ is converted by the machine, 0.94 kJ is lost in R_a and **8.43 kJ is returned to the supply** — regenerative braking (quadrant II).

Example 26.6 — Electric bike with a PMDC motor (Final Fall 2014 Q4)

Given. (p. 4.) No-load: 48 V gives 200 rpm (losses neglected). Locked rotor: 6 V gives 12 A and 27.5 N·m. **Find** (a) R_a and the machine constant; (b) on 48 V with a road-friction torque of 5.5 N·m: starting current and torque, steady current, speed and efficiency; (c) dynamic braking resistor to hold 24 km/h down an 11.54° slope (80 kg, g = 10 m/s², wheel diameter 600 mm, friction neglected). **Solution.** (a) Locked: $E_a = 0 \Rightarrow R_a = 6/12 = 0.5\ \Omega$. No load: $K\Phi = 48/20.94 = 2.292$ V·s/rad; from the torque test $27.5/12 = 2.292$ N·m/A — consistent.

b. Start: $I = 48/0.5 = 96$ A, $T = 220$ N·m. Steady: $I = 5.5/2.292 = 2.40$ A; $E_a = 48 - 1.2 = 46.8$ V ⇒ $n = 195$ rpm; η = $46.8/48 = 97.5$ % (only armature loss modelled). The speed approaches steady state exponentially with mechanical time constant $\tau = JR_a/(K\Phi)^2$ (0.19 s for an assumed J = 2 kg·m²).
c. Wheel speed $\omega = v/r = 6.667/0.3 = 22.2$ rad/s (212 rpm); driving torque from gravity = $mg\sin\theta\, r = 80(10)(0.2)(0.3) = 48.0$ N·m. Generator current $I = 48.0/2.292 = 20.95$ A; $E_a = 2.292(22.2) = 50.9$ V; $R_{ext} = E_a/I - R_a = 1.93\ \Omega$. (d, concept) With a capacitor instead of a resistor, the generated energy charges the capacitor (regenerative braking); the current falls as the capacitor voltage approaches E_a, and the stored charge can later drive the motor uphill.

Example 26.7 — Sudden voltage reduction: motoring to regeneration (Final Fall 2015 Q4)

Given. (pp. 11–12.) Separately excited motor, $R_a = 0.5\ \Omega$, no-load 1500 rpm at 220 V, friction negligible; constant load 100 N·m. **Find** (II) E_a; V_t, I_a, T_e, P_o at 1500 rpm with the load; (III) immediately after V_t drops to 200 V (speed momentarily unchanged, $K_a\Phi$ = 1.4): I_a, power flow, operating mode, subsequent response. **Solution.** (II) $E_a = 220$ V; $K_a\Phi = 1.401$; $I_a = 100/1.401 = 71.4$ A; $V_t = 220 + 35.7 = 255.7$ V; $T_e = 100$ N·m; $P_o = 15.7$ kW.

III. $I_a = (200 - 1.4 \times 157.08)/0.5 = -39.8$ A — reversed: the machine generates, $E_aI_a = -8.76$ kW flows back to the source; $T_e = -55.8$ N·m opposes rotation, and together with the load torque decelerates the shaft (both torques braking). As speed falls, E_a falls, the current swings back positive, and the motor settles where $I_a = 71.4$ A again: $E_a = 200 - 35.7 = 164.3$ V ⇒ 1121 rpm.

Example 26.8 — Restoring generator voltage by field or speed (Final Fall 2015 Q3 Part I)

Given. (p. 9.) 800 rpm, 220 V at no load with I_f = 0.47 A through R_f = 470 Ω (from a 220 V field supply); $R_a = 0.3\ \Omega$; load 60 A. Assume $E_a \propto I_f$. **Find** (a) V_t; (b) R_f to restore 220 V; (c) speed to restore 220 V with R_f = 470 Ω. **Solution.** (a) $V_t = 220 - 18 = 202$ V. (b) Need $E_a = 238$ V ⇒ $I_f = 0.47(238/220) = 0.508$ A ⇒ $R_f = 220/0.508 = 433\ \Omega$. (c) $n = 800(238/220) = 865$ rpm.

Example 26.9 — Converter-fed motor (Sen Example 4.11; link to Chapter V)

Given. 220 V separately excited motor, $R_a = 0.25\ \Omega$, $K_a\Phi$ = 0.18 V/rpm, 40 A, fed by a single-phase full converter from 265 V ac at firing angle 30°, current continuous and ripple-free. **Find** speed, torque and input power. **Solution.** $V_t = (2\sqrt{2} \times 265/\pi)\cos 30^\circ = 206.6$ V (Sen Eq. 4.58; controlled rectifiers are beyond EE361, which covers diode rectifiers in Lessons 30–31 — α = 0 gives the diode-bridge value 238.6 V); $E_a = 206.6 - 10 = 196.6$ V ⇒ 1092 rpm; $K_a\Phi = 0.18 \times 60/(2\pi) = 1.719$ N·m/A ⇒ $T = 68.8$ N·m; $P = V_tI_a = 8.26$ kW.

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7. Optional practice

P26.1. A separately excited motor (R_a = 0.4 Ω) runs at 1200 rpm drawing 30 A from 240 V. Find the speed when the load torque doubles, field constant.

▼ Solution
$E_a = 240 - 12 = 228$ V ⇒ $K_a\Phi = 228/125.66 = 1.814$. Double torque ⇒ 60 A ⇒ $E_a = 216$ V ⇒ $n = 1200(216/228) = 1137$ rpm.

P26.2. The same motor, at the original 30 A load, has its field reduced by 10 %. Find the new current and speed immediately and in steady state (torque constant).

▼ Solution
Steady state: T constant ⇒ $I_a = 30/0.9 = 33.3$ A; $E_a = 240 - 13.3 = 226.7$ V; $n = 226.7/(0.9 \times 1.814) \times 60/(2\pi) = 1325$ rpm (field weakening raises speed). Immediately (speed unchanged): $E_a = 0.9 \times 228 = 205.2$ V ⇒ $I_a = 34.8/0.4 = 87$ A — a current surge that accelerates the motor.

P26.3. Design a resistance starter for a 220 V motor with R_a = 0.25 Ω and rated current 40 A, keeping 40 A ≤ I_a ≤ 80 A. Would two steps be enough? How many are needed?

▼ Solution
Step 1: $R_1 + R_a = 220/80 = 2.75$ Ω ⇒ external 2.5 Ω; switch at 40 A when $E_a = 220 - 40(2.75) = 110$ V. Step 2: $(220 - 110)/80 = 1.375$ Ω ⇒ external 1.125 Ω; switch at $E_a = 220 - 55 = 165$ V. Step 3: $(220 - 165)/80 = 0.6875$ Ω ⇒ external 0.4375 Ω; at $E_a = 220 - 27.5 = 192.5$ V, cutting out all resistance gives $(220 - 192.5)/0.25 = 110$ A > 80 A, so a further step is needed: 0.344 Ω total (0.094 external), then at $E_a = 206.25$ V full voltage gives 55 A ✓. So two steps are not enough; four resistance steps are needed.

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8. Summary

Relation	Validity
$V_t = E_a + I_aR_a$ (motor), $V_t = E_a - I_aR_a$ (generator, I_a out)	steady state
$E_a = K_a\Phi\omega_m$, $T = K_a\Phi I_a$, $P_e = E_aI_a = T\omega_m$	electromagnetic
$\omega_m = V_t/(K_a\Phi) - R_aT/(K_a\Phi)^2$	separately excited motor, constant field
$I_{start} = V_t/(R_a + R_{ext})$	standstill ($E_a = 0$)
$E_a \propto \Phi\,\omega_m$; magnetization curve scales with speed	ratio method
$V_t < E_a$ while turning ⇒ regeneration	four-quadrant drives

9. Vocabulary

English	Türkçe
separately excited / shunt / series / compound	yabancı uyartımlı / şönt / seri / kompunt
terminal (external) characteristic	dış (uç) karakteristiği
armature reaction	endüvi reaksiyonu
rotational loss	dönme (sürtünme ve vantilasyon) kaybı
starting current; starter	kalkış akımı; yol verici
speed control; field weakening	hız kontrolü; alan zayıflatma

English	Türkçe
speed–torque characteristic	hız–moment karakteristiği
regenerative braking; dynamic braking	rejeneratif frenleme; dinamik (dirençli) frenleme
prime mover	tahrik makinesi (birincil hareket kaynağı)
self-excitation; critical field resistance	kendinden uyartım; kritik alan direnci

10. Sources

- LN15 viewer pp. 41–79 (slides 27–44). Notes.
- Sen (supplement): §4.2.7 pp. 139–141; §4.3.1–4.3.2 pp. 141–150 incl. Examples 4.2–4.3; §4.4.1 pp. 161–167 incl. Example 4.6; §4.4.3 pp. 176–180 incl. Example 4.10; §4.5 pp. 180–185 incl. Example 4.11 (viewer = printed + 24).
- Final Fall 2014 Q3–Q4 (pp. 3–4); Final Fall 2015 Q3–Q4 (pp. 9–12); Final Fall 2018 Q2 (pp. 4–5).
- Verification: `lesson25_26_dc_machines.py` (includes a start-up ODE and energy accounting for regenerative braking).
- More solved archive problems: Ch IV archive collection §2–6 (Final Fall 2017 Q3–Q4, Final Fall 2016 Q3–Q4, Final Fall 2018 Q3).

Chapter IV — DC-machine archive problems with full solutions

Assistant-written worked solutions, 2 November 2026. These final-exam questions complement the examples inside Lesson 25 and Lesson 26. Original papers are in `sources/exams/archive/Final/` *(page = PDF viewer page). Figures are restated, not copied; handwritten keys were read visually and every number recomputed. Numbers:* `assistant/calculations/full-semester/collection_ch4_archive.py` *(33 checks). Archive papers show past problem types, not the 2026 exam scope.*

Contents:

1. Final Fall 2017 Q2 Part II — EMF of a rotating coil, commutator, three coils (Lesson 25)
2. Final Fall 2017 Q3 — tower crane: holding, dynamic braking (Lesson 26)
3. Final Fall 2017 Q4 — closed-loop field control of a separately excited generator (Lesson 26)
4. Final Fall 2016 Q4 — traction vehicle: armature-voltage control, regeneration, field weakening (Lesson 26)
5. Final Fall 2018 Q3 — eDumper: regenerative downhill, motoring uphill, energy balance (Lesson 26)
6. Final Fall 2016 Q3 — long-compound generator (**extension**: compound machines are beyond the 2026 syllabus row “separately excited DC generators and motors”)
7. Final Fall 2018 Q4 — multiple-choice concepts across Chapters II–IV
8. Jan 2023 Final Q2 — moving core: saturation current, force and voltage along a motion profile (Lessons 22–23)
9. Jan 2023 Final Q3 — inductor and brushed DC machine structure (Lessons 16, 25)
10. Jan 2023 Final Q4(a)–(c) — crane motor: rated operation, holding, regenerative lowering with reversed field (Lesson 26; part (d) buck converter in Lesson 28)
11. Jan 2022 Final Q3 Part II — PM DC wind generator charging a battery through a diode (Lesson 26)

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1. Final Fall 2017 Q2 Part II — the simplest DC machine

Source. `exams/archive/Final/Final - Fall2017.pdf` pp. 6–7 (question), pp. 16–17 (key). **Key confirmed.**

(a) A coil of N turns rotates at constant speed ω in a uniform field; take θ = ωt with θ = 0 in the drawn position. The flux through the coil is the field times the projected area: $\phi(t) = \Phi_{max} \sin \omega t$, with $\Phi_{max} = B\, A_{coil}$. Faraday’s law with slip-ring terminals A–B:

$$e_{AB}(t) = -N\frac{d\phi}{dt} = -N\Phi_{max}\,\omega \cos \omega t.$$

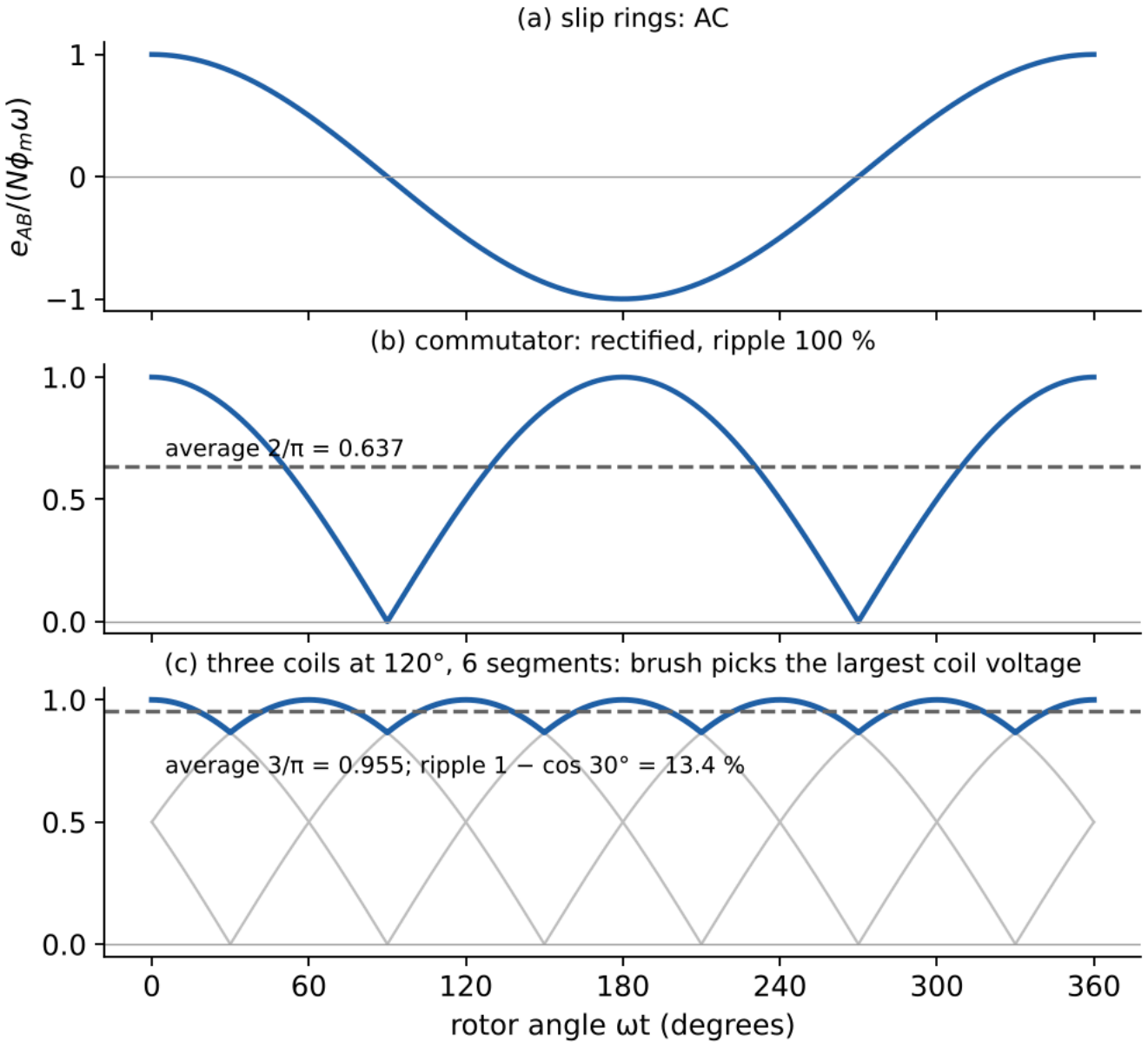
An AC voltage of amplitude $N\Phi_{max}\omega$, **proportional to speed**, at the rotation frequency (top panel of the figure; the sign depends on the chosen positive direction). Equivalently, each coil side moves with velocity v through B and $e = 2NB\ell v \cos \omega t$ (Lesson 25 §4.1).

(b) With a two-segment commutator, the brushes swap the coil connections every half turn — exactly when the coil voltage passes through zero. The terminal voltage is the rectified waveform

$$v_{AB}(t) = N\Phi_{max}\omega\,|\cos \omega t|,$$

average $\frac{2}{\pi}N\Phi_{max}\omega$ = 0.637 × peak, ripple 100 % (middle panel). The commutator is a mechanical full-wave rectifier (LN15 slides 17–24).

(c) Three identical coils 120° apart with six segments: each coil voltage is the same rectified sine shifted by 120°, and the brushes always connect the coil whose voltage is largest (the one under the pole centre). The output follows the **upper envelope**: six humps per revolution, minimum $\cos 30^\circ = 0.866$ of the peak, ripple $1 - \cos 30^\circ$ = 13.4 % of peak, average $\frac{3}{\pi}$ = 0.955 × peak (bottom panel). More coils and segments → smaller ripple and higher average; in a real armature many coils are connected in series in each parallel path so their voltages add (Lesson 25 §4.3).



2. Final Fall 2017 Q3 — separately excited motor in a tower crane

Source. `exams/archive/Final/Final - Fall2017.pdf` pp. 8–9 (question), pp. 18–19 (key). **Key confirmed** (it uses g = 10 m/s²).

Given. Field current constant at rated; friction and field loss ignored. R_a = 1 Ω, P_{rated} = 5 kW (output), V_{rated} = 225 V, I_{rated} = 25 A, N_{rated} = 1800 rpm.

(a) $E_a = V_t - I_a R_a = 225 - 25 = 200$ V at $\omega_m = 1800 \times 2\pi/60 = 188.5$ rad/s ⇒ $K_a\Phi = 200/188.5$ = **1.061 V·s/rad**. $T_{rated} = K_a\Phi I_a = 1.061 \times 25$ = **26.5 N·m**. Consistency: $E_a I_a$ = 5000 W = the rated output (no rotational loss).

(b) $\eta = P_{out}/P_{in} = 5000/(225 \times 25) = 5000/5625$ = **88.9 %**; the only loss is $I_a^2 R_a$ = 625 W.

(c) Holding the load at zero speed: $\omega_m = 0$ ⇒ $E_a = 0$. The required torque 26.5 N·m needs $I_a = 26.5/1.061$ = **25 A**, so $V_t = I_a R_a$ = **25 V**. All 625 W input is dissipated in R_a; holding a load with a DC motor costs only copper loss, but continuously.

(d) The motor is disconnected from the supply and an external resistor is connected across the armature; the load drives the machine as a generator and its energy is burned in the resistors — **dynamic braking** (Lesson 26 §4.4). At 900 rpm: $E_a = 1.061 \times 94.25$ = 100 V. Rated current (rated torque, enough to hold the load at constant speed): $I_a = E_a/(R_a + R_{ext})$ ⇒ $25 = 100/(1 + R_{ext})$ ⇒ R_{ext} = **3 Ω**.

(e) The drum radius/gearing is not given, so use energy: at constant speed the potential-energy rate equals the electrical dissipation, $mgv = I_a^2(R_a + R_{ext}) = 25^2 \times 4 = 2500$ W (also $= T\omega = 26.5 \times 94.25$). With m = 1000 kg and g = 10 m/s²: v = **0.25 m/s** (0.255 m/s with g = 9.81).

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3. Final Fall 2017 Q4 — generator with closed-loop terminal-voltage control

Source. `exams/archive/Final/Final - Fall2017.pdf` pp. 10–11 (question, Fig. 4), pp. 20–21 (key). **Key confirmed.**

Given. 25 kW, 1250 rpm, 250 V, 100 A separately excited generator; R_a = 0.5 Ω, R_f = 100 Ω. A PI controller adjusts the field rectifier so that V_t = 250 V from no load to full load. Magnetization curve at 1250 rpm (Fig. 4b): straight line from the origin to (2.2 A, 275 V), then a flatter line through (2.6 A, 300 V). Constant speed; friction and windage negligible.

Part I — no load. (a) I_a = 0 ⇒ $E_a = V_t$ = **250 V**. (b) On the linear segment: $I_f = 250 \times 2.2/275$ = **2.0 A**. (c) $V_f = I_f R_f$ = **200 V**.

Part II — full load (100 A). (d) $E_a = V_t + I_a R_a = 250 + 50$ = **300 V**. (e) From the curve, 300 V needs I_f = **2.6 A** (the controller raises V_f to 260 V). (f) $P_m = E_a I_a$ = **30 kW** converted from mechanical to electrical form. (g) $T_e = P_m/\omega_m = 30\,000/130.9$ = **229.2 N·m** (equivalently $K_a\Phi = 300/130.9 = 2.292$, $T = 2.292 \times 100$). (h) Including the field loss $2.6^2 \times 100$ = 676 W: $\eta = 25/(30 + 0.676)$ = **81.5 %** (83.3 % if the field supply is excluded). Losses: 5 kW in R_a (an unusually large 0.2 pu resistance), 0.68 kW field.

Interpretation. Without the controller the terminal voltage would fall from 250 V to 200 V at full load (the armature drop); the loop compensates by moving along the magnetization curve. Because the second segment is flatter (saturation), 20 % more field current buys only 9 % more EMF.

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4. Final Fall 2016 Q4 — traction vehicle with a separately excited motor

Source. `exams/archive/Final/Final - Fall2016.pdf` pp. 9–13 (question with Figs. 4.1–4.2, and two handwritten solutions, pp. 10–13). **Key confirmed.**

Given. Direct drive (no gearbox), wheel radius r = 0.5 m. Motor: rated V_t = 750 V, rated I_f = 2 A, R_a = 0.2 Ω, rated output 250 kW; magnetization $E_a = 375 I_f$ V at 382 rpm (linear). V_t from a bidirectional DC–DC converter (0–750 V); I_f from a separate converter (0–2 A). Friction neglected. Uphill the resisting force is F_d = 12.5 kN; downhill the net driving force (gravity) is 10 kN in the direction of motion.

Method. $K_a\Phi = E_a/\omega_m$ at the rated field: $750/(382 \times 2\pi/60) = 750/40.0$ **= 18.75 V·s/rad**. Steady state: $T_e = F\,r$, $I_a = T_e/K_a\Phi$, $\omega_m = (V_t - I_a R_a)/K_a\Phi$, i.e. the speed–torque line

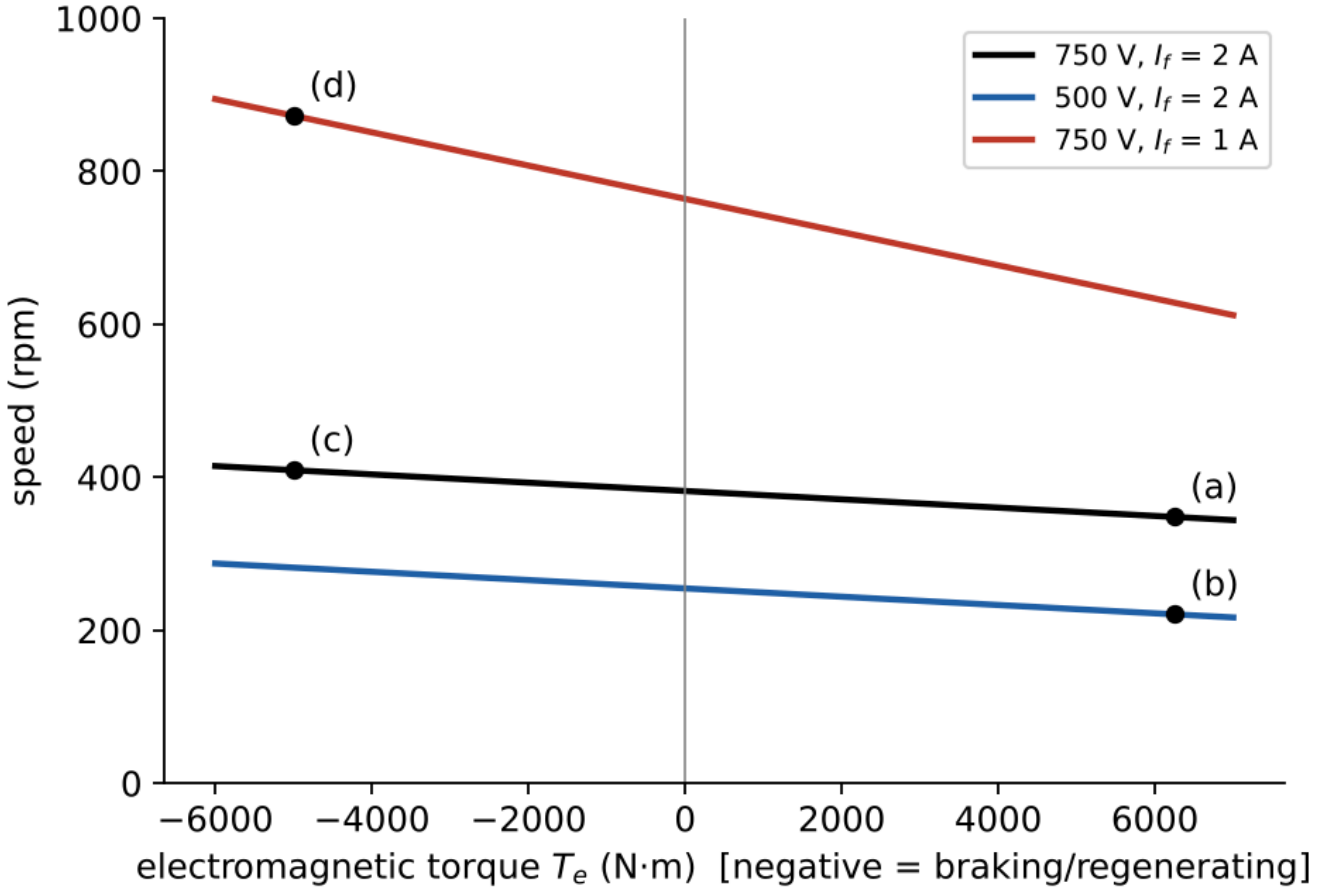
$$\omega_m = \frac{V_t}{K_a\Phi} - \frac{R_a}{(K_a\Phi)^2} T_e.$$

Case	V_t, I_f	T_e (N·m)	I_a (A)	E_a (V)	speed	vehicle
(a) uphill	750 V, 2 A	+6250	+333.4	683.3	36.45 rad/s = **348 rpm**	18.2 m/s = **65.6 km/h**
(b) uphill	500 V, 2 A	+6250	+333.4	433.3	23.11 rad/s = **220.7 rpm**	**41.6 km/h**
(c) downhill	750 V, 2 A	−5000	−266.7	803.3	42.85 rad/s = **409 rpm**	**77.1 km/h**
(d) downhill	750 V, 1 A	−5000	−533.4	856.7	91.38 rad/s = **873 rpm**	**164.5 km/h**

(a)–(b) Reducing the armature voltage moves the speed–torque line down parallel to itself: the same torque (same current, 333 A ≈ the rated 250 kW/750 V) at a lower speed. This is the wide-range speed control of Lesson 26 §4.3.

(c) Downhill the vehicle pushes the motor: the steady-state torque must oppose motion, T_e = −5000 N·m, so I_a reverses. $E_a = V_t + |I_a| R_a$ = 803 V > V_t: the machine is **generating** in the forward direction (quadrant II). With the bidirectional converter the power flows back to the battery — **regenerative braking**: shaft power 214 kW, returned at the terminals $V_t|I_a|$ = 200 kW, 14 kW lost in R_a.

(d) Halving the field halves $K_a\Phi$ (9.375 V·s/rad, linear magnetization). To produce the same braking torque the armature current must double (533 A, 1.6 × rated — an overload), and the no-load speed doubles: the steady downhill speed rises to 873 rpm (164.5 km/h, physically unrealistic for a bus). **Field weakening raises speed**; for braking it is the wrong direction — strengthen the field or lower V_t to brake harder at lower speed.



5. Final Fall 2018 Q3 — eDumper electric mining truck

Source. `exams/archive/Final/Final - Fall2018.pdf` pp. 7–9 (question with Figs. 3–5; blue handwritten key on pp. 8–9). **Key confirmed.**

Given. PM DC motor (constant flux), R_a = 6 mΩ, bidirectional DC–DC converter 91.6 % efficient each way, no friction, g = 10 m/s², slope 7.5°. Downhill loaded: 123 t. Uphill empty: 58 t.

(a) Downhill. V_t = 182 V, I_a = 3000 A at 600 rpm. The load drives the machine: generating mode, current flows *out* of the positive terminal. $E_a = V_t + I_a R_a = 182 + 18$ = **200 V**. Power flow: mechanical $E_a I_a$ = 600 kW in → $I_a^2 R_a$ = 54 kW loss → electrical $V_t I_a$ = 546 kW out to the converter. Motor efficiency (as generator) $\eta = V_t I_a / E_a I_a = 182/200$ = **91 %**.

(b) No friction, constant speed: the gravity component along the slope $F = mg \sin\alpha = 123\,000 \times 10 \times \sin 7.5^\circ$ = 160.5 kN does all the work. $P_{mech} = Fv \Rightarrow v = 600\,000/160\,547$ = 3.737 m/s = **13.45 km/h**. Energy into the battery in 12 min: $546 \times 0.916 \times (12/60)$ = **100.0 kWh** (the key writes 600 × 0.91 × 0.916 × 0.2 — the same product). Distance travelled: 3.737 × 720 = 2691 m.

(c) Uphill, empty, at 2 × 13.45 = 26.9 km/h = 7.474 m/s. Mechanical power $P = 58\,000 \times 10 \times \sin 7.5^\circ \times 7.474$ = **565.9 kW**. Same flux, twice the speed ⇒ $E_a = 2 \times 200$ = **400 V** (motoring now: current *into* the positive terminal). $I_a = P/E_a$ = **1415 A**; $V_t = E_a + I_a R_a = 400 + 8.5$ = **408.5 V**; $I_a^2 R_a$ = 12.0 kW. Battery power = $V_t I_a / 0.916$ = 630.9 kW ⇒ overall efficiency $565.9/630.9$ = **89.7 %** (= 0.916 × 0.979). *Can it climb back?* Same distance at twice the speed ⇒ 6 minutes. Battery energy used: 630.9 kW × 0.1 h = **63.1 kWh < 100 kWh** ✓. Physical reason: the climb lifts only the 58 t truck, while the descent was powered by the 123 t loaded truck over the same height;

even after two conversion chains (≈ 83 % down, ≈ 90 % up) there is energy left over — the eDumper idea. Bonus: accelerating to 7.47 m/s needs $\frac{1}{2}mv^2$ = 1.62 MJ = 0.45 kWh at the wheels (≈ 0.5 kWh from the battery) — negligible.

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6. Final Fall 2016 Q3 — cumulatively compounded long-shunt generator (extension)

Source. `exams/archive/Final/Final - Fall2016.pdf` pp. 7–8 (question with circuit and piecewise magnetization; handwritten key). **Key confirmed.** *Extension:* the 2026 syllabus lists separately excited machines only; this problem is included because it is recoverable and reuses the same KVL + magnetization-curve method (Sen §4.3.2–4.3.3).

Given. R_a = 0.8 Ω, series field R_s = 0.2 Ω (in the armature branch: *long* shunt), shunt field R_f = 350 Ω across the terminals, 1500 rpm. Shunt-field contribution: E_a = 6 V for I_f < 0.01 A; $E_a = (180/0.3)I_f$ for 0.01 ≤ I_f < 0.3 A; $E_a = 150 + 100I_f$ for $I_f \geq 0.3$ A. Series field adds $1.4I_a$ (the hint allows the two EMF contributions to be added).

Circuit. $V_t = R_f I_f$, $I_a = I_L + I_f$, $E_a - I_a(R_a + R_s) = V_t$, $E_a = E_{sh}(I_f) + 1.4I_a$.

(a) Open circuit. $I_L = 0 \Rightarrow I_a = I_f$ and $E_a = I_f(R_a + R_s + R_f) = 351I_f$. On the upper segment: $351I_f = 150 + 100I_f + 1.4I_f \Rightarrow I_f$ = 0.6010 A, E_a = 210.9 V, $V_t = 350I_f$ = **210.3 V**. The lower segments give no other solution ($601.4I_f = 351I_f$ only at zero; on the residual segment $6 + 1.4I_f = 351I_f$ needs I_f = 0.017 A, outside its range), so the voltage builds up from the 6 V residual to this single operating point.

(b) R_L = 5 Ω. $V_t = 350I_f = 5I_L \Rightarrow I_L = 70I_f$, $I_a = 71I_f$. KVL: $71I_f(1.0) + 350I_f = 150 + 100I_f + 1.4(71I_f) \Rightarrow 421I_f = 150 + 199.4I_f \Rightarrow I_f$ = 0.6769 A. Then I_a = 48.06 A, E_a = 285.0 V, V_t = **236.9 V**. The terminal voltage *rises* with load (cumulative compounding: the series field more than compensates the I_aR_a drop) — "over-compounded".

(c) $P_{out} = V_t^2/R_L$ = 11.23 kW. Losses: shunt field $I_f^2R_f$ = 160 W; armature + series field $I_a^2(R_a + R_s)$ = 2310 W; friction and windage 150 W; total 2620 W. $\eta = 11.23/(11.23 + 2.62)$ = **81.1 %**. Check: input = $E_aI_a + 150$ = 13.85 kW ✓.

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7. Final Fall 2018 Q4 — multiple choice across Chapters II–IV

Source. `exams/archive/Final/Final - Fall2018.pdf` pp. 10–12. The key circles answers for every item except 4.7, whose page area is blank in the supplied scan; our answer to 4.7 is marked. Each item lists the reasoning, because the exam gives zero for any wrong mark.

Item	Answer	Reasoning
4.1 Losses in a rotating conductive steel disc between stationary PMs; steel B–H single-valued	**B, D** (key)	Flux in each part of the moving disc changes → eddy currents in the conductive steel; friction/windage of the rotating disc. No coil → no copper loss; the given steel curve has no loop → no hysteresis loss. (Lesson 16)

Item	Answer	Reasoning
4.2 Magnetic materials	**E** (key)	A: electrical steel has high μr and B_sat (≈1.5–1.8 T) for 50 Hz. B: ferrites have very low conductivity → tiny eddy loss at high f. C: eddy loss ∝ f²B². D: a distributed gap (powder core) has less fringing than one discrete gap. (Lessons 11, 13, 16)
4.3 Three-phase transformers in Turkey	**B, D** (key)	Distribution is 50 Hz, 230 V line-to-neutral (400 V LL), which needs a neutral: Δ–Δ has none → A false. Δ–Y (Dyn11) is the standard step-down distribution connection → C false. A three-phase bank rated "34.5 kV/230 V" is specified by line-to-line voltages, so its LL ratio is 34.5k/230 — the *turns* ratio is different (Lesson 21 §3.2).
4.4 Zero regulation	**A** (key)	VR = 0 when $\lvert V_1\rvert = \lvert V_2'\rvert$: the tip of $\mathbf{V}_1$ lies on the circle of radius V_2' (dashed arcs). That happens in Load 1 with a leading current (Lesson 20 §2.4).
4.5 Initial motion when current is applied	**A, D** (key)	Torque $= \frac{1}{2}i^2 dL/d\theta$ (singly excited) or $i_s i_r\, dM/d\theta$. System 1: salient rotor between aligned and unaligned → $dL/d\theta \neq 0$ → torque. System 2: round rotor, uniform gap → no torque. System 3: salient rotor already aligned → $dL/d\theta = 0$ (equilibrium). System 4: round rotor carrying its own current, coil axis at an angle → mutual torque. (Lesson 24)
4.6 Induced voltage in a brushed DC machine	**A, C, D** (key)	A: commutator rectification gives a DC component. B false: a motor has a back EMF. C: PM field cannot be varied. D: the magnetization curve saturates. (Lesson 25)
4.7 Using at least two of the three torque components $\frac{1}{2}i_s^2 dL_s/d\theta$, $\frac{1}{2}i_r^2 dL_r/d\theta$, $i_s i_r\, dL_{sr}/d\theta$	**D** (ours; key area blank)	A: round–round, both excited → only the mutual term. B: round–round, stator only → no torque. C: salient rotor, stator only → only the reluctance term. D: salient rotor, both excited → reluctance + mutual (Lesson 24 §4.3, Example 24.3 = Final 2014 Q2).

Item	Answer	Reasoning
4.8 Series-wound DC machine	**B** (key)	$I_f = I_a \Rightarrow T \propto I_a^2$ (unsaturated). A false: it can generate (rarely used, unstable voltage). C false: field and armature are the same circuit. D false: it gives very high torque at low speed (traction). (Sen §4.4.3 — extension)
4.9 Why laminate	**B** (key)	Laminations break eddy-current loops (loss ∝ thickness²); they do not change hysteresis loss or μr and make the core less robust, not more. (Lesson 16)

8. Jan 2023 Final Q2 — moving core: force and induced voltage along a motion profile

Source. `exams/archive/20221/EE361_Final_2022_Q4_Solution.pdf` pp. 6–9 (the file is the **16 January 2023** final despite its name; no key for Q2 in the file). Lessons 22, 23.

Given (restated from the figure). A stationary T-shaped core (a vertical bar of width W with a middle leg of width $2W$ carrying N = 100 turns) and a moving C-shaped part (back leg and two arms of width W). The moving part is pulled to the left by F_{ext}; three equal gaps of length x open: between the middle leg and the back leg (area $2W \times L$) and between each arm end and the bar (area $W \times L$ each). W = 10 mm, L = 50 mm (depth), H = 50 mm. Steel B–H: vertical line (μ → ∞) up to saturation at 1.6 T. Motion: $x = (1 + 1000t)$ mm for 0–4 ms, 5 mm for 4–8 ms, $(13 - 1000t)$ mm for 8–12 ms. Leakage, fringing, friction, core loss and winding resistance ignored.

Magnetic circuit. Gap 1 ($\mathcal{R}_1 = x/\mu_0 2WL$) in series with gaps 2 ∥ 3 ($x/\mu_0 WL$ each, parallel $x/\mu_0 2WL$): $\mathcal{R} = x/(\mu_0 WL)$ and

$$L(x) = \frac{N^2 \mu_0 W L}{x} = \frac{6.283 \times 10^{-6}}{x}\ \text{H} \quad (x \text{ in m}): \qquad L(1\ \text{mm}) = 6.283\ \text{mH},\ L(5\ \text{mm}) = 1.257\ \text{mH}.$$

The middle-leg flux splits equally between the two arms, so B is the same in all three gaps: $B = \Phi/(2WL) = \mu_0 N i/(2x)$.

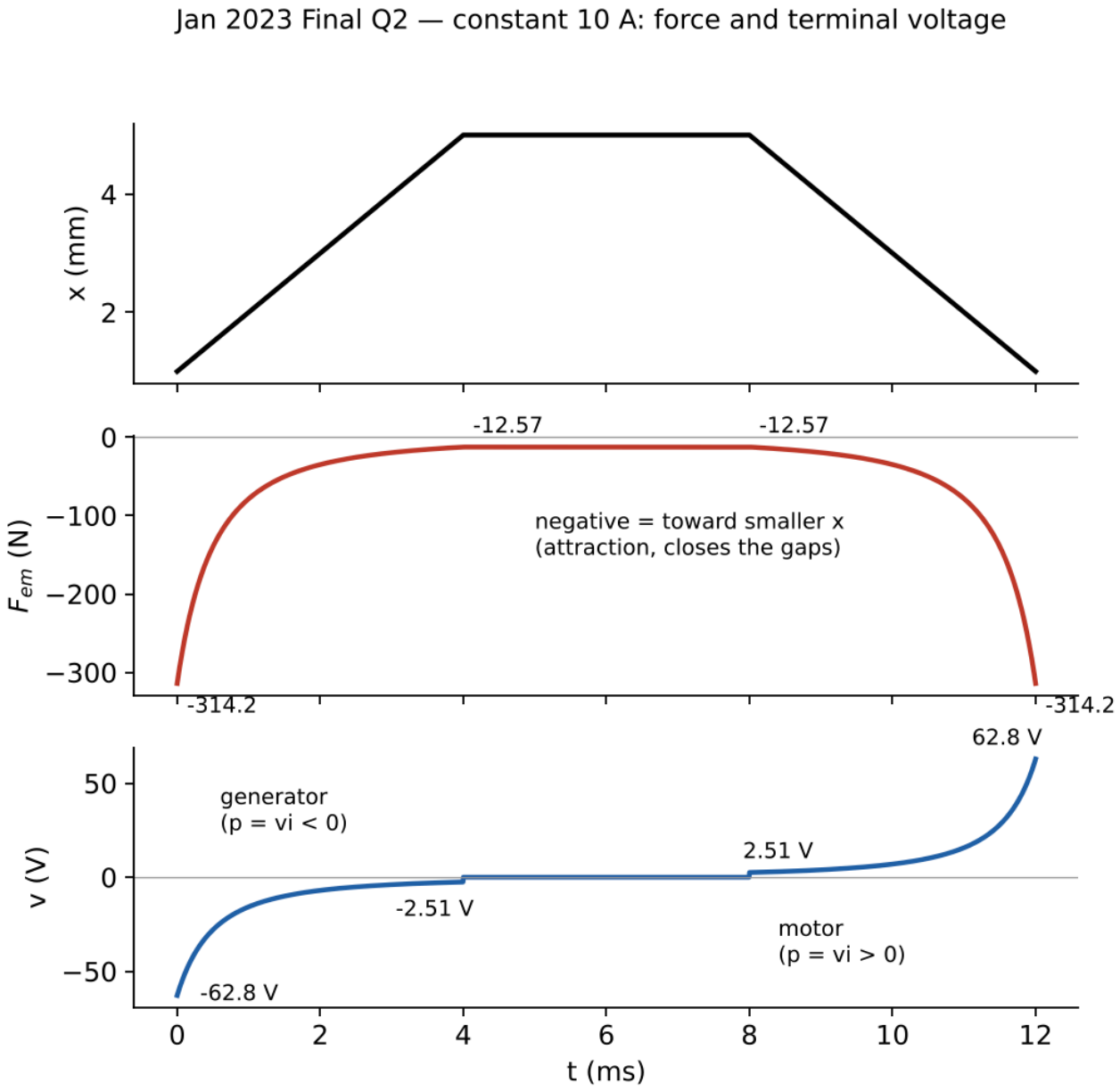
(a) Saturation first occurs where B is largest for a given current, i.e. at the **smallest gap, x = 1 mm** (at 0 and 12 ms). $i_{max} = 2xB_{sat}/(\mu_0 N) = 2(10^{-3})(1.6)/(4\pi \times 10^{-7} \times 100)$ **= 25.5 A**.

(b) At constant i = 10 A: $F_{em} = \frac{1}{2} i^2\, dL/dx = -\frac{1}{2}(100)(6.283 \times 10^{-6})/x^2 = -3.14 \times 10^{-4}/x^2$ N. The minus sign means the force acts toward **smaller x** (the parts attract). Values: **−314.2 N at 0 ms**, **−12.57 N at 4 ms and 8 ms** (constant in between), **−314.2 N at 12 ms**; on the ramps $F(t) = -314.2/(1 + 1000t)^2$ N (0–4 ms, t in s) and its mirror image on 8–12 ms. Check by magnetic stress: at 1 mm, B = 0.628 T over a total pole area of 2000 mm² gives $B^2 A/2\mu_0$ = 314 N ✓.

(c) With R = 0 and constant current, $v = d(Li)/dt = i\,\dfrac{dL}{dx}\dfrac{dx}{dt} = -\dfrac{i\,k\,\dot{x}}{x^2}$ ($k = 6.283 \times 10^{-6}$ H·m; the quotient-rule hint). $\dot{x}$ = +1 m/s, 0, −1 m/s on the three intervals:

- 0–4 ms: from **−62.8 V** (just after 0) to **−2.51 V** (just before 4 ms);
- 4–8 ms: **0 V** (no motion, constant L, constant i);
- 8–12 ms: from **+2.51 V** (just after 8 ms) to **+62.8 V** (just before 12 ms). The voltage jumps at 4 ms and 8 ms because the velocity jumps; at those instants quote both one-sided values.

(d) 0–4 ms: the gap opens against the magnetic pull (the external force does work) and $p = vi < 0$ — electrical energy flows *out* of the winding: **generator**. 8–12 ms: the magnetic force closes the gap and does positive work, $p = vi > 0$: **motor**. 4–8 ms: no motion, no power: neither. Energy over 0–4 ms (numerically integrated): electrical energy delivered to the source 0.503 J = 0.251 J of mechanical work done against F_{em} + 0.251 J released from the field ($\frac{1}{2}i^2\Delta L$) — the constant-current "half and half" rule of Lesson 22 §5.3.



9. Jan 2023 Final Q3 — inductor and brushed DC machine structure

Source. same file, pp. 10–11 (no key). Concept answers; drawings are described in words.

Part 1 (inductor). (a) A closed core (e.g. E–E or toroid) of high-permeability material with an N-turn winding on one leg (usually the centre leg), often with an air gap in the centre leg. (b) Equivalent circuit: series winding resistance R_w (finite conductor resistivity) feeding a parallel combination of the magnetizing inductance L_m and a core-loss resistance R_c (eddy currents from finite core conductivity plus hysteresis from the non-zero loop area) — Lesson 16 §4.2; leakage

ignored as allowed. (c) The gap: makes $L \approx N^2\mu_0 A_g/l_g$ insensitive to the steel's μ (temperature, tolerance); raises the saturation current; stores most of the energy (Ch II–III collection §3 shows 98 %); linearizes the λ–i characteristic.

Part 2 (brushed DC machine). (a) Stator yoke with salient field poles carrying the field winding; rotor (armature) core with slots holding the armature winding; commutator and brushes on the shaft. (b) For **counter-clockwise motoring** with the stator N pole on the left and S pole on the right (field from left to right across the air gap): $\mathbf{F} = I\boldsymbol{\ell} \times \mathbf{B}$ must push the conductors under the N pole downward and those under the S pole upward, so armature conductors under the **N pole carry current into the page (⊗)** and those under the **S pole out of the page (⊙)**. The field winding current on each pole circulates so that, by the right-hand rule, the left pole face is a north pole. Reversing either the field or the armature current reverses the rotation. (c) Table:

Part	Rotating / stationary	Purpose
Field winding (on stator poles)	stationary	create the stationary magnetic field (given)
Commutator	rotating	mechanical rectifier: reverses each coil's current as it passes the neutral zone so torque keeps one direction
Armature winding	rotating	carries the load current; site of EMF and torque production
Brush	stationary	sliding contact between the external circuit and the commutator
Stator yoke / pole cores	stationary	low-reluctance return path for the field flux
Armature (rotor) core and shaft	rotating	carry the flux across the rotor, hold the conductors, transmit torque

10. Jan 2023 Final Q4(a)–(c) — crane motor with armature and field control

Source. same file, pp. 12–14 (blue handwritten key). **Key confirmed**, with one direction remark (E17). Part (d) — supplying the field from 800 V with a buck converter, D = 0.375 — is solved in Chapter V, Lesson 28.

Given. Separately excited motor: rated V_t = 800 V, 955 rpm, I_a = 25 A, R_a = 3.2 Ω, R_f = 200 Ω, rated I_f = 1.5 A.

(a) Lifting at rated speed. $E_a = 800 - 3.2(25)$ = 720 V; ω = 955 × 2π/60 = 100.0 rad/s ⇒ $K_a\Phi$ = 7.20 V·s/rad. Output (no rotational loss) $P_{out} = E_a I_a$ = **18.0 kW**; $T = P/\omega$ = **180 N·m**. Input = armature 800×25 + field $1.5^2 \times 200$ = 20 000 + 450 = 20.45 kW ⇒ η = **88.0 %**.

(b) Holding 90 N·m at zero speed. E_a = 0; $I_a = 90/7.2$ = 12.5 A; $V_t = I_a R_a$ = **40 V**.

(c) Lowering at 160 rad/s with V_t = 800 V and $I_f = -0.75 \times 1.5$ A. Signs: take lifting rotation and the lifting torque as positive. Now $K_a\Phi = -0.75 \times 7.2 = -5.40$ V·s/rad and $\omega = -160$ rad/s, so $E_a = (-5.40)(-160)$ = **+864 V** > V_t ⇒ generating. $I_a = (800 - 864)/3.2$ = **−20 A** (current leaves the

positive terminal). $T = K_a \Phi I_a = (-5.40)(-20)$ = **+108 N·m**. Power flow: mechanical in 108×160 = 17.3 kW = $E_a |I_a|$; $I_a^2 R_a$ = 1.28 kW lost; **16.0 kW returned to the 800 V supply**; the field source still supplies $1.125^2 \times 200$ = 253 W. *Direction:* the torque is positive — the **same direction as the lifting torque in (a)** and opposite to the present downward rotation, which is exactly what holds a descending load at constant speed. The key's note "(opposite direction compared to part a)" should be read as "opposite to the rotation" (E17). Why reverse the field? With V_t fixed at +800 V and the rotation reversed, E_a must stay positive and exceed 800 V to regenerate; reversing Φ restores the sign of E_a, and weakening it to 75 % sets the speed at which E_a = 864 V.

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11. Jan 2022 Final Q3 Part II — PM DC wind generator charging a battery

Source. `exams/archive/Final/EE361_2022_Final_Exam_Q4_solution.pdf` p. 10 (31 January 2022 final; no key for Q3). Lesson 26 §3.

Given. PM brushed DC machine, K = 1.25 V·s/rad; ideal diode in series with a 24 V battery; charging power 384 W at 200 rpm; rotational loss 25 W at all speeds; R_a unknown.

Find R_a first. At 200 rpm, ω = 20.94 rad/s, $E_a = 1.25 \times 20.94$ = 26.18 V; battery current $384/24$ = 16 A ⇒ $R_a = (26.18 - 24)/16$ = **0.136 Ω**.

(d) 960 W ⇒ I = 40 A ⇒ $E_a = 24 + 40(0.136)$ = 29.45 V ⇒ $\omega = 29.45/1.25$ = 23.56 rad/s = **225 rpm**.

(e) Electromagnetic torque $KI = 1.25 \times 40$ = 50 N·m; rotational-loss torque $25/23.56$ = 1.06 N·m ⇒ wind torque **51.1 N·m**. Power check: 51.06×23.56 = 1203 W = 960 (battery) + 218 ($I^2 R_a$) + 25 ✓.

Remarks. The diode blocks reverse current, so the machine never motors from the battery; charging starts only when E_a > 24 V, i.e. above the cut-in speed $24/1.25$ = 19.2 rad/s = 183 rpm. Between cut-in and 225 rpm the charging power rises from 0 to 960 W — very steeply, because $I = (K\omega - 24)/R_a$ and R_a is small.
